

	FIR VEHICLE TESTING		05/06/2023
FIR No :Testing-Outdoor/516800/2023/6074			
Part No/ Group :51685253R			
Subject :Report on fluid economy evaluation of TML Signa 3525.T Cx Atulya 6.7L (250 HP engine) OBD-II BS-VI vehicle with G950 (FGR: 9.36) Transmission and 5.12 Rear axle ratio in comparison with TML Signa 3518.T BS-VI OBD-I Base vehicle.			
Test Vehicle	Project Name :Signa 3525.T Cx Atulya 6.7L (250 HP engine) OBD-II BS-VI vehicle Model Name : Signa 3525.T Cx BS-VI OBD-II		Enclosures : Annexure 1 - 6
	Chasis Number :PC73550122RJ00815 Engine Number :22H64232488		
FIR Status :	ESO FIR		

Role	Name	Signature
Prepared By	Palash Das	 05-Jun-2023
Checked By	Swapnil Jadhav	 05-Jun-2023
Approved By	Ajaykumar Jadhav	 05-Jun-2023

Cc: Head CVBU Engg, Chief Engineer, Head COC, Project Team [Design]; Project Team [Testing]

PA 20 - F 222 / May19

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Documents : Please click here

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 FIR/Testing-Outdoor/516800/2023/6074	FIR VEHICLE TESTING	Date: 03/06/2023		
Part No / Group: 51685253R	Specification no :- " 259051680013 "		Action by	Failure class

Subject: -	Report on fluid economy evaluation of TML Signa 3525.T Cx Atulya 6.7L (250 HP engine) OBD-II BS-VI vehicle with G950 (FGR: 9.36) Transmission and 5.12 Rear axle ratio in comparison with TML Signa 3518.T BS-VI OBD-I Base vehicle.			
1.0 Conclusion: 1.1 Fluid economy in rated load condition at 40 kmph : - Fluid economy of Signa 3525.T Cx BS-VI OBD-II vehicle was found 3.67 kmpl which was Superior by 5.97% wrt Signa 3518.T BS-VI OBD-I Base vehicle. - Fluid economy of Signa 3525.T Cx BS-VI OBD-II vehicle (Without THU Bearing) was found 3.60 kmpl which was Superior by 3.85% wrt Signa 3518.T BS-VI OBD-I Base vehicle. 1.2 Fluid economy in rated load condition at 55 kmph : - Fluid economy of Signa 3525.T Cx BS-VI OBD-II vehicle found 3.47 kmpl which was superior by 5.18% wrt Signa 3518.T BS-VI OBD-I Base vehicle. - Fluid economy of Signa 3525.T Cx BS-VI OBD-II vehicle (Without THU Bearing) was found 3.40 kmpl which was Superior by 3.08% wrt Signa 3518.T BS-VI OBD-I Base vehicle. 1.3 Fluid economy in Unladen condition at 40 kmph : - Fluid economy of Signa 3525.T Cx BS-VI OBD-II vehicle found 6.4 kmpl which was superior by 3.38% wrt Signa 3518.T BS-VI OBD-I Base vehicle.		--		--
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1.4 <u>Fluid economy in Unladen condition at 55 kmph :</u> - Fluid economy of Signa 3525.T Cx BS-VI OBD-II vehicle found 5.9 kmpl which was which was Superior by 5.04% wrt Signa 3518.T BS-VI OBD-I Base vehicle. 1.5 <u>Overall fluid economy status at 40 kmph (80% Laden + 20% unladen condition):</u> - Overall Fluid economy of Signa 3525.T Cx BS-VI OBD-II vehicle found 5.57 kmpl which was Superior by 4.17% wrt Signa 3518.T BS-VI OBD-I Base vehicle. - Overall Fluid economy of Signa 3525.T Cx BS-VI OBD-II vehicle (Without THU Bearing) was found 5.54 kmpl which was Superior by 3.53% wrt Signa 3518.T BS-VI OBD-I Base vehicle. - Signa 3525.T Cx BS-VI OBD-II meets the PALS target of "Leadership". 1.6 <u>Overall fluid economy status at 55 kmph (80% Laden + 20% unladen condition):</u> - Overall Fluid economy of Signa 3525.T Cx BS-VI OBD-II vehicle found 5.17 kmpl which was Superior by 5.08% wrt Signa 3518.T BS-VI OBD-I Base vehicle. - Overall Fluid economy of Signa 3525.T Cx BS-VI OBD-II vehicle (Without THU Bearing) was found 5.14 kmpl which was Superior by 4.44% wrt Signa 3518.T BS-VI OBD-I Base vehicle. - Signa 3525.T Cx BS-VI OBD-II meets the PALS target of "Leadership".	--	--
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2.0 Result: Vide Observations		
FAILURES CLASS: A: Failure that happen without warning. Detrimental to vehicle / personnel, road users and safety. Can cause death / hospitalization. Noncompliance of regulatory requirement. B: Vehicle gets crippled with sufficient warning. Minor injury to occupant & road users (First aid only). Premature failures / repeated failures. Leads to consequential catastrophic damages. Need immediate attention. C: Customer irritants. Capable of damaging aesthetics. Failure can be attended at next servicing Schedule. D: Customer perception issues and minor irritants.		
3.0 Test Vehicles: - 1. Signa 3525.T BS-VI OBD-II (ERCJ-1256)_Test vehicle Chassis no. - PC73550122RJ00815; Engine no. - 22H64232488. 2. Signa 3518.T BS-VI OBD-I (ERCJ 1160)_Base vehicle Chassis no. - PC53564012RJ00688; Engine no. - 12C63863032. Input Doc Ref.: - Test Request No: - Testing-Outdoor/516800/2023/6074 DID No: – NA. DDP No: – NA.	Enclosures: Annexure 1: Panel chart Annexure 2: PAT sheet Annexure 3: Duty cycle report Annexure 4: Test request Annexure 5: Test plan Annexure 6: EOL	
Work Order No. :- P.22.EA.501	Make :- TATA Motors Limited	T I No. – 41616
Earlier Ref: - Nil.		
4.0 Project Status:- DR3		
5.0 Introduction: As a part of RDE (OBD-II) Testing Program, TML Signa 3525.T Cx Atulya 6.7L BS-VI OBD-II vehicle with G950 (FGR:9.36) Transmission and 5.12 Rear axle ratio was offered for fluid economy evaluation along with TML Signa 3518.T BS-VI OBD-I as a base vehicle. All vehicles were tested back-to-back at JSR- Ranchi-JSR route in laden and unladen conditions at 40 km/hr as well as 55 km/hr vehicle cruising speeds. This report will reveal detail of this trial with the status.		
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6.0 Testing:

Following test was conducted during evaluation :

Sr. no.	Test	Reference standard	Instruments used
1	Fluid economy evaluation	TST/VT/WG/158	Cummins Data Logger

7.0 Vehicle details:

Parameter		Base Vehicle	Test vehicle
Vehicle model		Signa 3518.T (OBD-I) (ERC-1160)	Signa 3525.T (OBD-II) (ERC-1256)
A) Vehicle specification			
Engine Model		Cummins Ashwamedh 5.6L BS-VI (180 HP) OBD-I	Cummins Atulya 6.7L BS-VI (250 HP) OBD-II
Max Power		180 HP @2300 RPM	250 HP @2300 RPM
Max Torque		850 Nm @ 1000 ~1600 rpm	950 Nm @ 1000 ~1800 rpm
GVW (kg)		35000	35000
Gear Box		Model: Tata G950, Type: DD	Model: Tata G950, Type: DD
Gear Ratios.		1 st gear:9.36, 2 nd gear: 5.2, 3 rd gear: 3.05, 4 th gear: 1.84, 5 th gear: 1.33, 6 th gear: 1.00	1 st gear:9.36, 2 nd gear: 5.2, 3 rd gear: 3.05, 4 th gear: 1.84, 5 th gear: 1.33, 6 th gear: 1.00
Axle Ratio		5.12	5.12
Tyres	FA1	Size: 295/90R20, Make: M/s Apollo (Endurance) , Type: Regular (Non-LCRR)	Size: : 295/90R20, Make: M/s Bridgestone, Type : LCRR (Phase-II)
	FA2	Size: 295/90R20, Make: M/s Apollo (Endurance MA) , Type: Regular (Non-LCRR)	Size: : 295/90R20, Make: M/s Bridgestone, Type : LCRR (Phase-II)
	LA1	Size: 295/90R20, Make: M/s Apollo (Endurance LD) , Type: Regular (Non-LCRR)	Size: : 295/90R20, Make: M/s Bridgestone, Type : LCRR (Phase-II)
	RA1	Size: 295/90R20, Make: M/s Apollo (Endurance LD) , Type: Regular (Non-LCRR)	Size: : 295/90R20, Make: M/s Bridgestone, Type : LCRR (Phase-II)
	RA2	Size: 295/90R20, Make: M/s Apollo (Endurance LD) , Type: Regular (Non-LCRR)	Size: : 295/90R20, Make: M/s Bridgestone, Type : LCRR (Phase-II)
Tractive effort @ 1 st gear		74851 N	83657 N
Tractive effort @ Top gear		7997 N	8938 N
Fan Details (Type/Make/Blades)		Type: E-Viscous fan and Size:24"	Type: E-Viscous fan and Size:24"

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Part No / Group: 51685253R	Specification no :- " 259051680013 "	Action by Failure class

Parameter	Base Vehicle	Test vehicle
Vehicle model	Signa 3518.T (OBD-I) (ERC-1160)	Signa 3525.T (OBD-II) (ERC-1256)
B) FE Avenues _Engine Level		
Base Cal Improvement (Cal Maturity %)	As per production released	95% Cal Maturity
Low Viscous Engine Oil (TMGO-10W30)	No (15W40CK4+)	Yes (10W30)
FE Switch	Yes (3 Mode : E/B/P)	Yes (3 Mode : E/B/P)
Engine Features (EOL : Enable/Disable)	Enable	Enable
ATS Optimization	Yes	Yes
APM	No	Yes
Cruise Control	No	No
Gear based VAM	No	No
Intelligent mode shifter	No	No
Moving Mass Estimation	No	No
C) FE Avenues _Vehicle Level		
Driveline Optimization	NA	Yes
GSA (Gear Shift Advisor)	Yes	Yes (Not Tuned)
MSA (Mode Shift Advisor)	NA	No
Low CRR Tyres	No	Yes
Low Viscous rear axle oil (80W90)	Yes	Yes
Smart Alternator (LIN)	NA	No
THU/Unitised/Low friction bearing	No	Yes
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8.0 Test result:

8.1 Cruising speed: 40 Km/hr & 55 Km/hr :

Sr. No.	Parameters	Base Vehicle : OBD-1				Test Vehicle : OBD-2			
		SIGNA 3518.T Atulya BSVI				SIGNA 3525.T RDE BSVI			
1	Vehicle Identification No. (ERC No./EJ No./Reg. No.)	ERC-1150				ERC-1203(Released) ERC-1256(Iteration 1)			
2	Vehicle GVW/GCW	34600				34600			
3	Engine Details (Make/cz/Norms/No. of Cyl.)	Cummins (5B6/ Ashwamedh				Atulya (6.7 Lit) B5 VI			
4	Engine Max Power	180 HP @ 2300 rpm				250 hp @ 2300 rpm			
5	Engine Max Torque	850 Nm @ 1000-1600 rpm				950 Nm @ 1000-1800 rpm			
6	Engine Calibration Maturity Level	As per production released				95% Cal Maturity			
7	Regeneration Strategy (Time use/Soot base)	Time Based (100 Hrs)				Time Based (100 Hrs)			
8	Gear Box (make/drive/FR)	TATA 6-990 00				TATA 6-990 00			
9	Gear Ratios	1-9.36, 2- 5.20, 3- 3.05, 4- 1.84, 5- 1.33, 6- 1.00				1-9.36, 2- 5.20, 3- 3.05, 4- 1.84, 5- 1.33, 6- 1.00			
10	RAR	5.12				5.12			
11	Type Details	295/90 R 20 RIB Apollo Non-LCRR				295/90R 20 RIB Bridgestone Low LCRR			
12	LBSR	High & Low Engine Speed Breakpoint				1700 rpm, 1700 rpm			
13	Vehicle Mass Threshold	kg				35000			
14	VAM	Variable Acceleration 1, 2				1.5, 1.5			
15	Variable Acceleration Speed 1, 2	kmph/sec				10, 40			
16	GDP	GD Max. Veh. Speed (Heavy & Light Engine Load)				45, 45			
17	RSG	Maximum Accelerator Vehicle Speed				80			
18	Torque Factor	Torque Limit Medium Adjustment Factor				100%			
19	Powertrain Protection (Enable / Disable)	Balance				95%			
20	PTP	Torque Limit				100%			
21	USFE / Torque Table Revision (1/2/3/4)	NA				Disable			
22	Fan Details (Type/Make/Blades)	USFE-1				24" High Efficiency Viscous fan			
23	Base Cal Improvement (Cal Maturity %)	E-24" High Efficiency Viscous fan				95% Cal Maturity			
24	FE Switch	As per production released				Yes (15W40C04)			
25	Engine Features (EOL : Enable/Disable)	No (15W40C04)				Yes (3 Mode : E/B/P)			
26	ATS Optimization	Yes (3 Mode : E/B/P)				Enable			
27	APM	Enable				Yes			
28	Cruise Control	No				No			
29	Gear based VAM	No				No			
30	Intelligent mode shifter	No				No			
31	Moving Mass Estimation	No				No			
32	DriveTime Optimization	No				Yes			
33	GSA (Gear Shift Advisor)	Yes				Yes (Not Tuned)			
34	MSA (Mode Shift Advisor)	No				No			
35	Low CRB Types	No				Yes			
36	Low Viscous rear axle oil (RDW90)	Yes				Yes			
37	Smart Alternator (LIN)	No				No			
38	THU/Unitted/Low friction bearing	No				Yes			
39	Route	Jamshedpur-Ranchi-Jamshedpur				Jamshedpur-Ranchi-Jamshedpur			
40	FE Duty Cycle (Laden/Unladen)	Unladen Trial @ 40 Kmph				Unladen @ 40 Kmph			
41	Iterations	Laden Trial @ 40 kmph				Laden @ 40 kmph			
42	Trial Date	Unladen @ 55 kmph				Unladen @ 55 kmph			
43	Weather Condition (Sunny/Rainy/Cloudy/Windy)	Laden @ 55 Kmph				Laden @ 55 Kmph			
44	Traffic Condition (Light/Average/Heavy)	Unladen @ 40 Kmph				Laden @ 40 kmph			
45	Regeneration (Yes/No)	Unladen @ 55 kmph				Unladen @ 55 kmph			
46	Trip Distance	Laden @ 55 Kmph				Laden @ 55 Kmph			
47	Total Diesel Quantity	Unladen @ 40 Kmph				Laden @ 40 kmph			
48	Total Urea Quantity	Unladen @ 55 kmph				Unladen @ 55 kmph			
49	Equivalent Urea Quantity : 50% Consumption	Laden @ 55 Kmph				Laden @ 55 Kmph			
50	Total Eq. Fluid Consumed	Unladen @ 40 Kmph				Laden @ 40 kmph			
51	Urea uses % wt. Diesel	Unladen @ 55 kmph				Unladen @ 55 kmph			
52	Fuel Economy	Laden @ 55 Kmph				Laden @ 55 Kmph			
53	Status of Fuel economy	Unladen @ 40 Kmph				Laden @ 40 kmph			
54	Equivalent Fluid Economy	Unladen @ 55 kmph				Unladen @ 55 kmph			
55	Status of Equivalent Fluid economy	Laden @ 55 Kmph				Laden @ 55 Kmph			
56	Cluster Total Odo. Reading (at End of trial)	Unladen @ 40 Kmph				Laden @ 40 kmph			
57	Cluster Trip AFE	Unladen @ 55 kmph				Unladen @ 55 kmph			
58	Status of Cluster Trip AFE wrt Actual Fluid economy	Laden @ 55 Kmph				Laden @ 55 Kmph			
59	Average Fuel Economy	Unladen @ 40 Kmph				Laden @ 40 kmph			
60	Status of Average Fuel economy	Unladen @ 55 kmph				Unladen @ 55 kmph			
61	Average Eq. Fluid Economy	Laden @ 55 Kmph				Laden @ 55 Kmph			
62	Average Eq. Fluid Economy (Extrapolated for Non-THU bearing)	Unladen @ 40 Kmph				Laden @ 40 kmph			
63	Status of Average Eq. Fluid economy	Unladen @ 55 kmph				Unladen @ 55 kmph			
64	Status of Average Eq. Fluid economy (Extrapolated for Non-THU bearing)	Laden @ 55 Kmph				Laden @ 55 Kmph			
65	Overall Avg. Fuel Economy (kmpl) : L + UL	Unladen @ 40 Kmph				Laden @ 40 kmph			
66	Status of Overall Avg. Fuel Economy (%)	Unladen @ 55 kmph				Unladen @ 55 kmph			
67	Overall Avg. Eq. Fluid Economy (kmpl) : L + UL	Laden @ 55 Kmph				Laden @ 55 Kmph			
68	Overall Avg. Eq. Fluid Economy (kmpl) : L + UL (Extrapolated for Non-THU bearing)	Unladen @ 40 Kmph				Laden @ 40 kmph			
69	Status of Overall Avg. Eq. Fluid Economy (%)	Unladen @ 55 kmph				Unladen @ 55 kmph			
70	Status of Overall Avg. Eq. Fluid Economy (%) (Extrapolated for Non-THU bearing)	Laden @ 55 Kmph				Laden @ 55 Kmph			

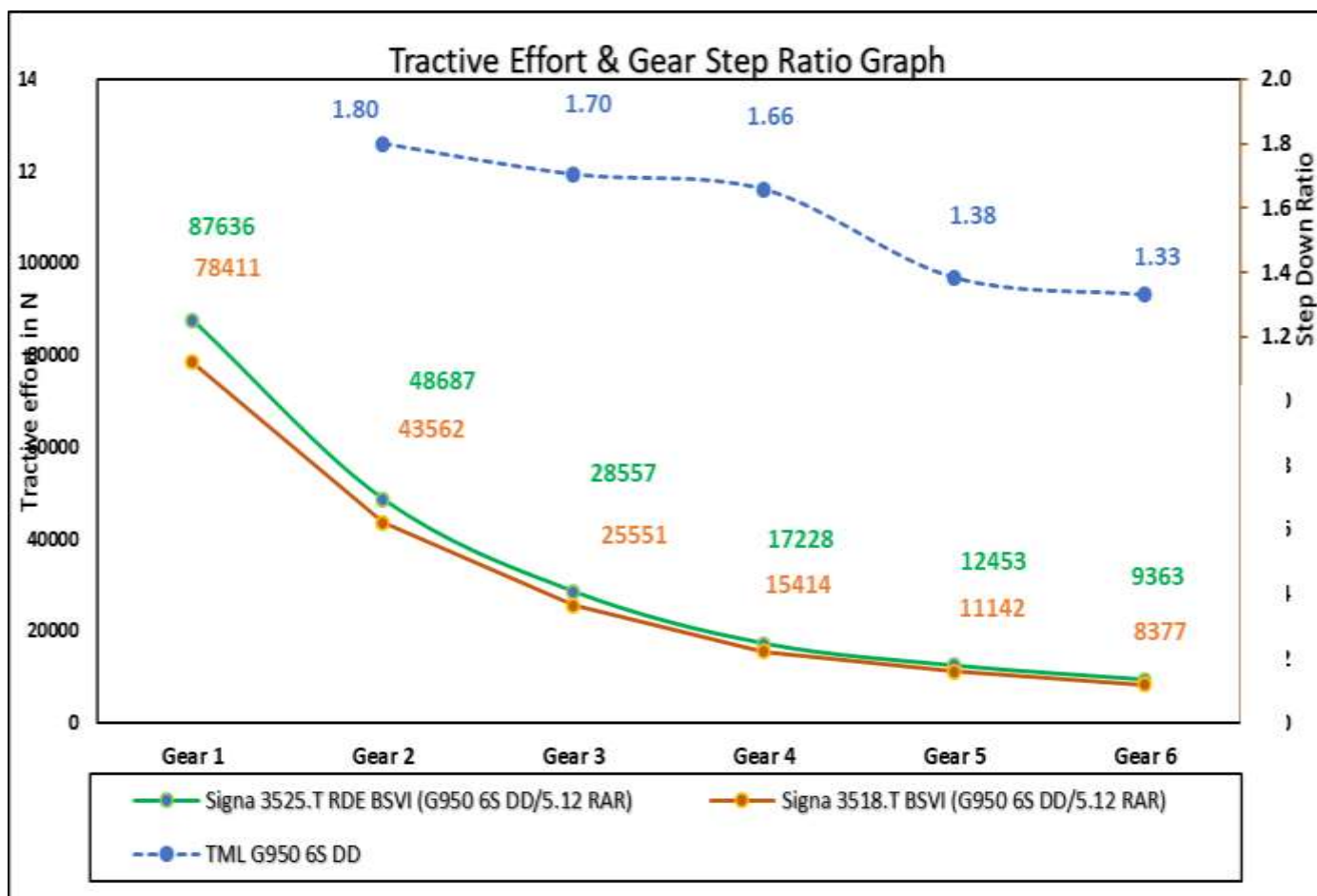
Note: Equivalent Urea consumption is taken as 50% of actual urea consumption.

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8.3 Tractive Effort Comparison :**Remarks:**

Signa 3525.T Cx BS-VI OBD-II (250 HP) Test vehicle tractive effort was better in all gears as compared with Signa 3518.T BS-VI OBD-I (180 HP) _ Base vehicle.

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8.4 Torque Table :

8.3.1 : Base vehicle (Signa 3518.T BS-VI OBD-I (180 HP))

➤ Economy mode

	600	800	1000	1200	1400	1600	1800	2000	2200	2400	2600	2800
1	500	500	850	753	400	400	400	400	400	400	80	0
1.33	500	500	850	800	753	543	400	400	400	400	80	0
1.84	500	500	850	800	753	543	400	400	400	400	80	0
3.05	500	500	850	800	800	753	543	400	400	400	80	0
5.2	450	500	850	850	850	850	772	694	616	412	80	0
9.36	450	500	850	850	850	850	772	694	616	412	80	0

➤ Balance mode

	600	800	1000	1200	1400	1600	1800	2000	2200	2400	2600	2800
1	500	500	850	850	753	543	400	400	400	400	80	0
1.33	500	500	850	850	800	753	543	400	400	400	80	0
1.84	500	500	850	850	800	753	543	400	400	400	80	0
3.05	500	500	850	850	800	800	753	543	400	400	80	0
5.2	450	500	850	850	850	850	772	694	616	412	80	0
9.36	450	500	850	850	850	850	772	694	616	412	80	0

8.3.2: Test vehicle (Signa 3525.T BS-VI OBD-II (250 HP))

➤ Economy mode

	600	800	1000	1200	1400	1600	1800	2000	2200	2400	2600	2800
1	800	800	950	650	495	495	495	450	417	396	119	0
1.33	800	800	950	650	554	554	554	513	463	440	119	0
1.84	800	800	950	650	554	554	554	513	463	440	119	0
3.05	800	800	950	684	606	606	606	513	463	440	119	0
5.2	800	800	950	950	950	950	950	890	809	580	119	0
9.36	800	800	950	950	950	950	950	890	809	580	119	0

➤ Balance mode

	600	800	1000	1200	1400	1600	1800	2000	2200	2400	2600	2800
1	800	800	950	950	735	655	655	572	510	503	119	0
1.33	800	800	950	950	735	655	655	572	510	503	119	0
1.84	800	800	950	950	735	655	655	572	510	503	119	0
3.05	800	800	950	950	735	655	655	516	510	503	119	0
5.2	800	800	950	950	950	950	950	890	809	580	119	0
9.36	800	800	950	950	950	950	950	890	809	580	119	0

9.0 Observation: Nil

10.0 Further Testing: Further fuel economy evaluation needs to be done with benchmark vehicle to get the PALS attribute wrt core competitor.

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Fuel Economy Evaluation on Signa 3525.T RDE BSVI with Signa 3518.T BSVI Base Vehicle.

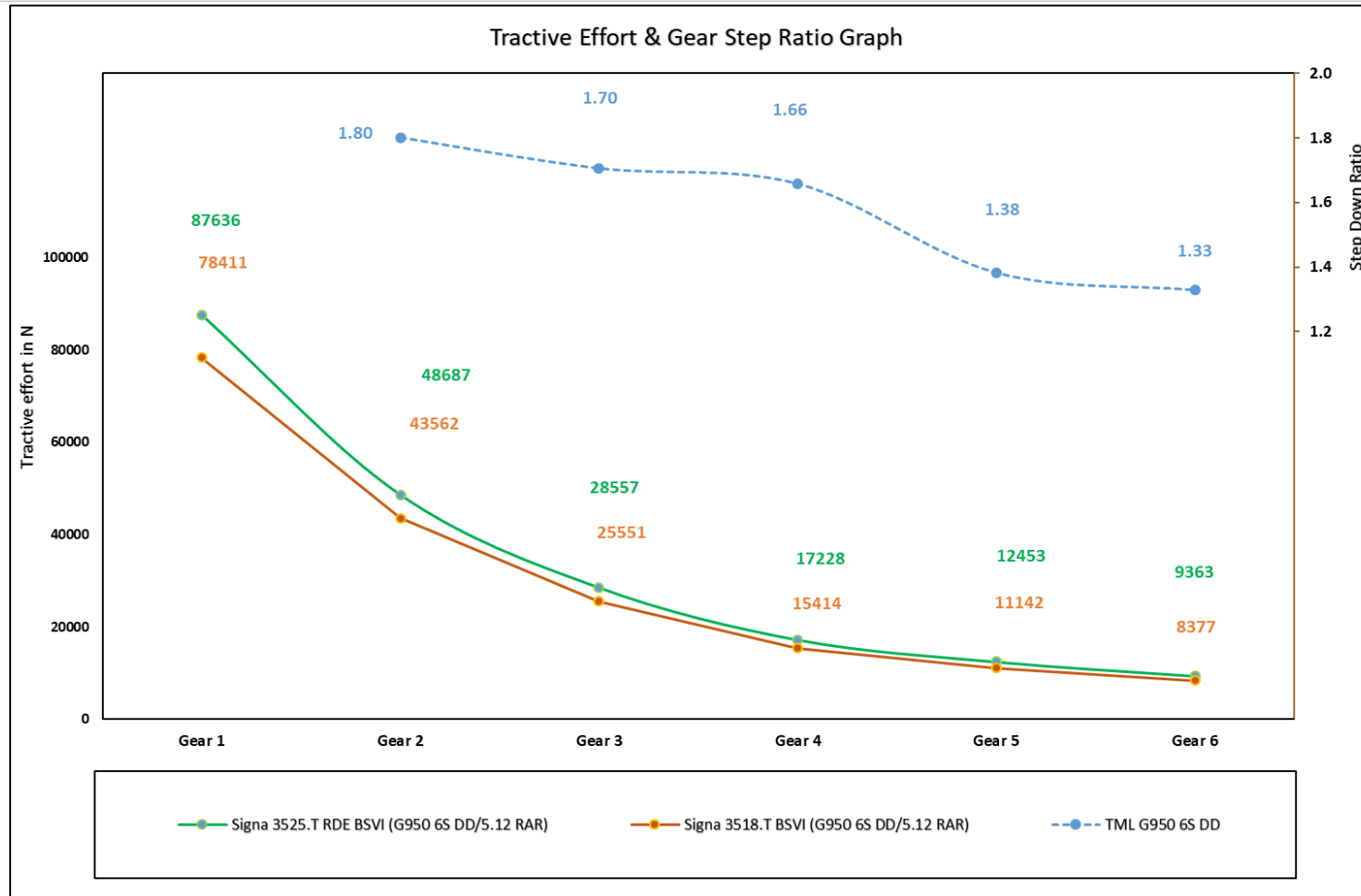
PED - HCV | 26.04.2023



Vehicle Specification

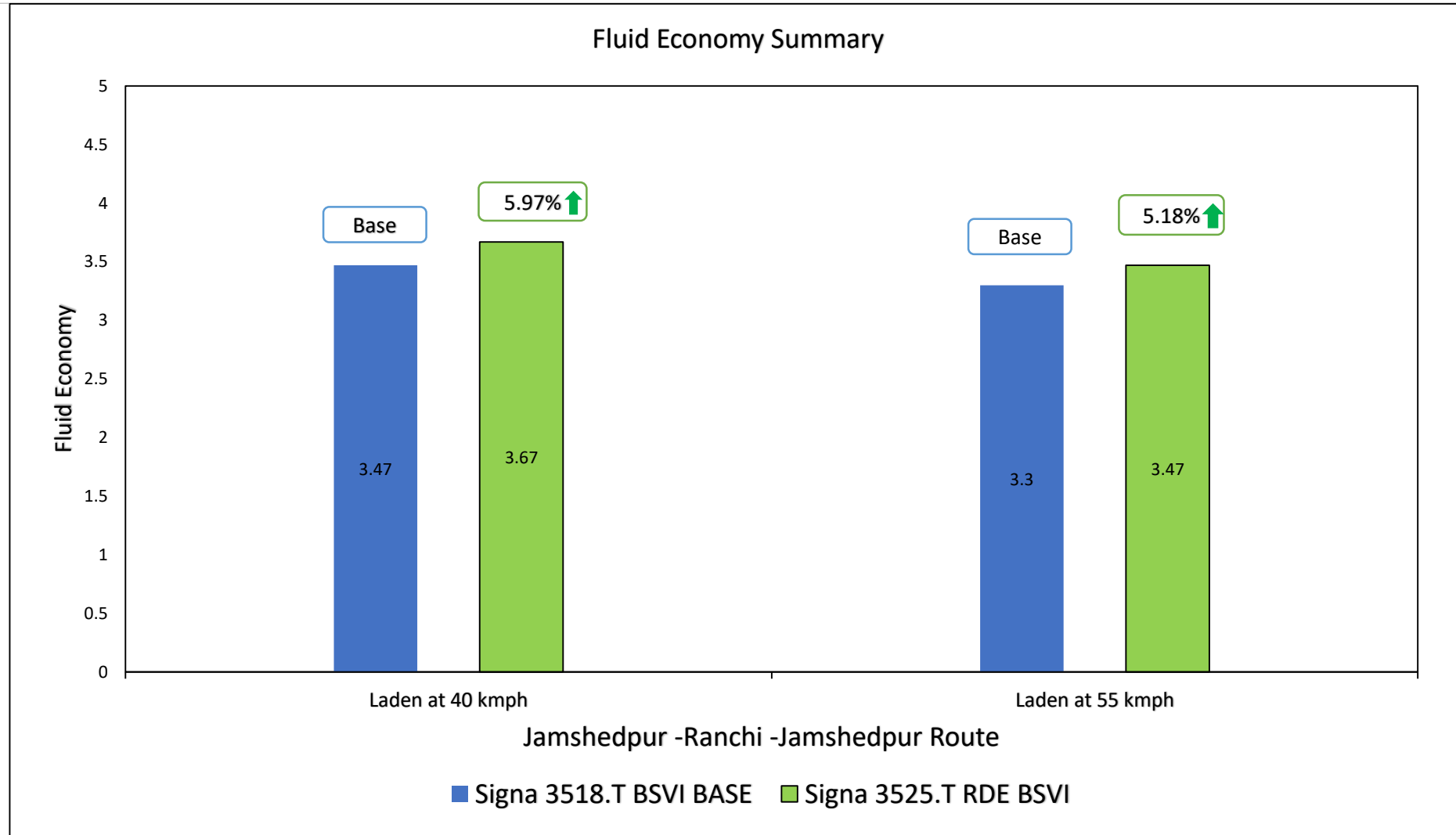
Specification		Test Vehicle	Base Vehicle
Vehicle	Model	Signa 3525.T Atulya BSVI RDE	Signa 3518.T Ash BSVI
	GVW (Rated load) Kg	35000(Kg)	35000(kg)
	GVW (Unladen) kg	10,680	10,320
Engine	Emission norm	BSVI RDE	BSVI
	Engine	Atulya BSVI RDE 6.7 Ltr	Ashwamedh BSVI 5.6L
	Calibration	95%	As per production released
	Technology	--	--
	Torque	950 Nm @ 1000-1800 rpm	850 Nm @ 1000 - 1600 rpm
	Power	250 HP @ 2300 rpm	180 HP @ 2300 rpm
	Torque to weight ratio	27.14	24.28
	Power to weight ratio	7.14	5.14
	Torque band	800	600
Driveline	Gear box	G950 6S DD, FGR- 9.36	G950 6S DD, FGR- 9.36
	Gear 1	9.36	9.36
	Gear 2	5.2	5.2
	Gear 3	3.05	3.05
	Gear 4	1.84	1.84
	Gear 5	1.33	1.33
	Gear 6	1	1
	RAR	5.12	5.12
	Tyre	295/90R20	295/90R20

Tractive Effort Comparison

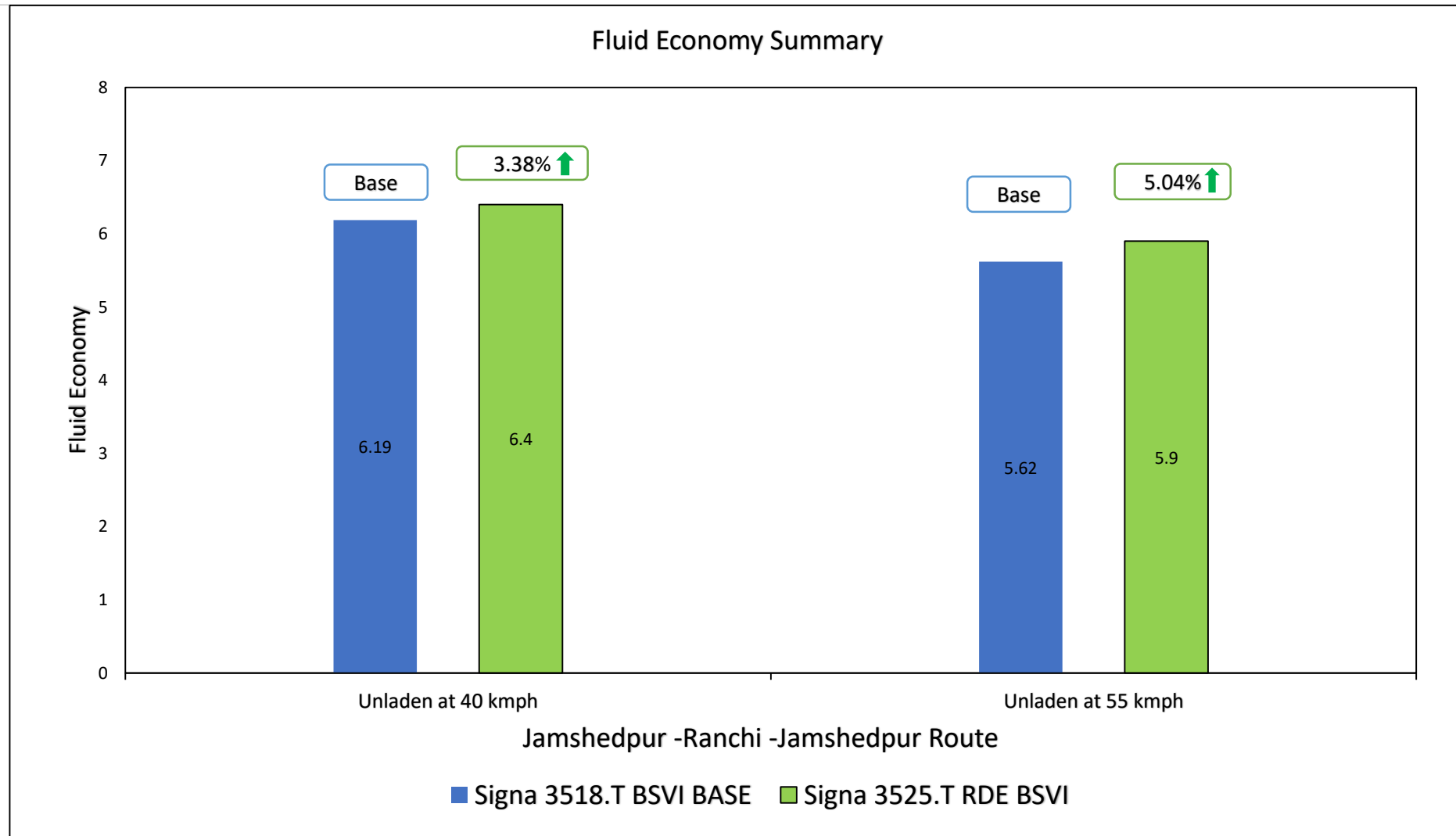


➤ Tractive effort of Signa 3525.T RDE vehicle is better than Signa 3518.T BSVI vehicle.

Summary FE Status with all FE Avenues(Laden)



Summary FE Status with all FE Avenues(Unladen)



Duty Cycle Analysis Jamshedpur-Ranchi Road FE Trial

Duty Cycle : Summary (Laden)

JSR-RANCHI-JSR Duty Cycle

		23.02.2023(40 kmph)		24.02.2023(55 kmph)	
Parameters		Signa 3525.T RDE Test Vehicle	Signa 3518.T BSVI Base Vehicle	Signa 3525.T RDE Test Vehicle	Signa 3518.T BSVI Base Vehicle
Engine idle speed (rpm)		650	750	650	750
Low idle time (%)		2.76	4.43	5.89	6.25
Average engine speed (rpm)		1,062.6	1,090.2	1282.48	1316
Average power (kW)		37.19	38.58	49.47	50.97
Avg. vehicle speed (kmph)	Including Stop	37.34	37.55	45.67	46.3
	Excluding Stop	39.44	39.42	48.84	49.49
FE switch utilization (%time)	Power	17.81	12.74	14.47	12.24
	Balance	0.9	15.07	12.26	24.49
	Eco	81.29	72.19	73.28	63.27
Gear utilization (%time)	Neutral Gear	5.4	3.8	6.6	6.6
	4 th Gear	3.3	4.5	2.8	3.2
	5 th Gear	7.69	9.97	7.17	6.93
	6 th Gear	79.41	75.34	78.75	77.45
	Total shifts	211	266	167	150
Throttle (%time)	100%	9.6	24.7	19.8	41.8
	0%	35.3	33.1	29.1	32.9
Coolant Temp (°C)	80°C	0.2	14.1	0.2	8
	85°C	1.7	46.9	1.5	52.7
	90°C	52.2	23.8	59.1	27
SCRTM Fuel spent (%)		--	--	--	--
Regeneration Fuel spent (%)		--	--	--	--
Duty Cycle fuel economy (kmpl)		4.11	3.95	3.95	3.78
Duty cycle Distance(km)		245.27	245.75	245.06	245.45
Duty Cycle Time(Hrs)		6.57	6.55	5.37	5.3
Actual Avg. Fuel Economy		3.78	3.57	3.61	3.42

Duty Cycle : Summary (Unladen)

JSR-RANCHI-JSR Duty Cycle

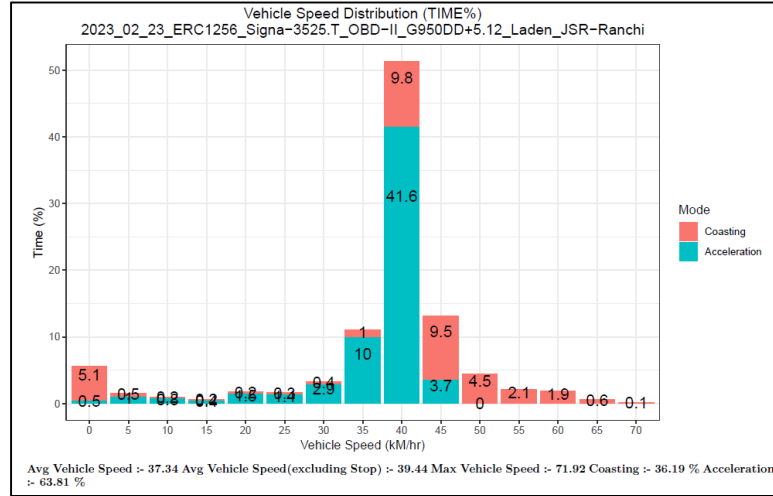
		02.03.2023(40 kmph)		03.03.2023(55 kmph)	
Parameters		Signal 3525.T RDE Test Vehicle	Signal 3518.T BSVI Base Vehicle	Signal 3525.T RDE Test Vehicle	Signal 3518.T BSVI Base Vehicle
Engine idle speed (rpm)		650	750	650	750
Low idle time (%)		3.3	3.04	6.38	5.39
Average engine speed (rpm)		1012.92	981.87	1,297.43	1,296.28
Average power (kW)		17.24	19.03	26.12	28.28
Avg. vehicle speed (kmph)	Including Stop	37.13	38.04	47.63	48.13
	Excluding Stop	38.59	39.25	51.41	50.94
FE switch utilization (%time)	Power	0.00	0.00	1.09	0.00
	Balance	0.00	0.00	0.47	0.00
	Eco	100	100	98.44	100
Gear utilization (%time)	Neutral Gear	4	3.3	7.5	6
	4 th Gear	1.5	1.4	1.1	1.5
	5 th Gear	2.3	1.4	1.8	1.7
	6 th Gear	88.3	91.0	86.73	87.76
	Total shifts	124	90	92	75
Throttle (%time)	100%	0.1	1.1	0.4	2.9
	0%	11.9	14.9	13.8	18.4
Coolant Temp (°C)	80°C	0.3	14.4	0.9	2.4
	85°C	1.9	82.4	1.3	90.1
	90°C	93.7	1.7	93.2	4.3
SCRTM Fuel spent (%)		--	--	--	--
Regeneration Fuel spent (%)		0.00	0.00	0.00	0.00
Duty cycle fuel economy (kmpl)		6.97	7.13	6.9	6.61
Duty cycle Distance(km)		242.33	221.41	242.53	213.44
Duty Cycle Time(Hrs)		6.53	5.82	5.09	4.43
Actual Avg. Fuel Economy		6.6	5.79	6.07	6.37

Vehicle Speed Distribution

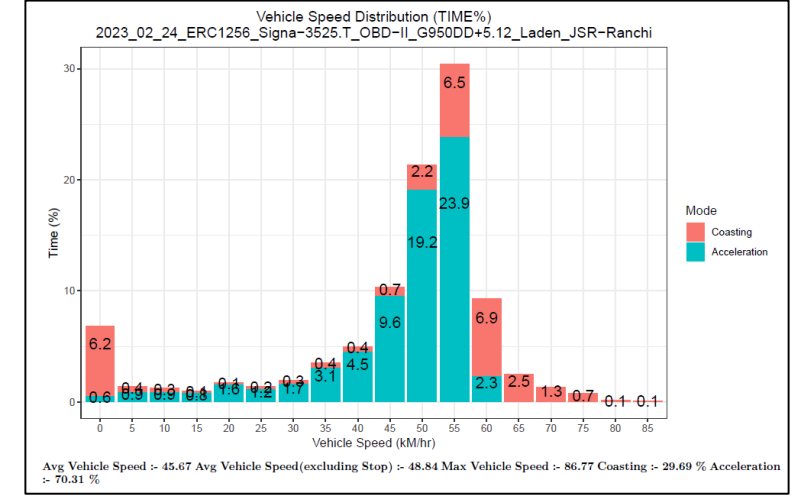
Laden Trial

Signa 3525.T RDE Test Vehicle

23.02.23 40 kmph

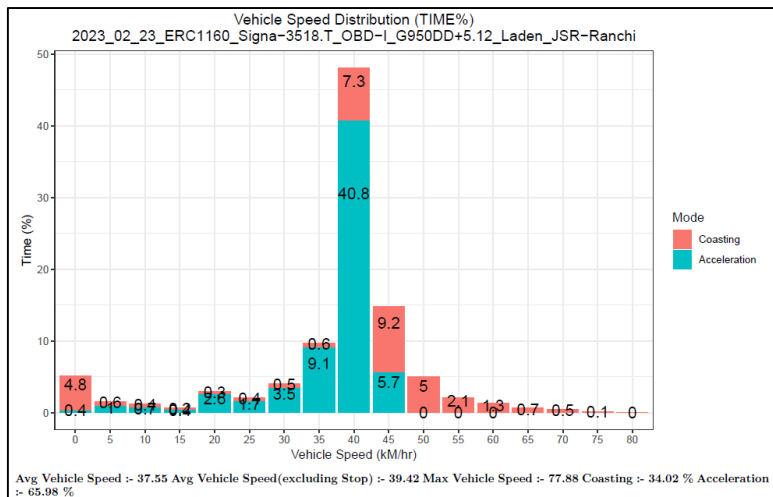


24.02.23 55 kmph

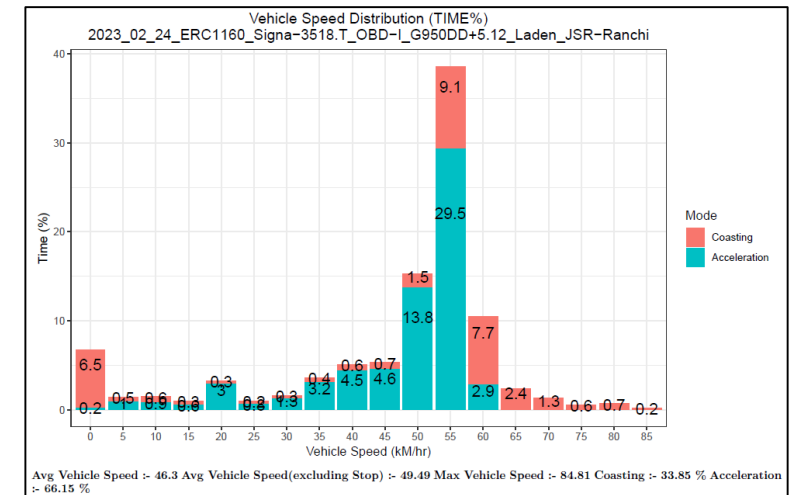


Signa 3518.T Base Vehicle

23.02.23 40 kmph



24.02.23 55 kmph

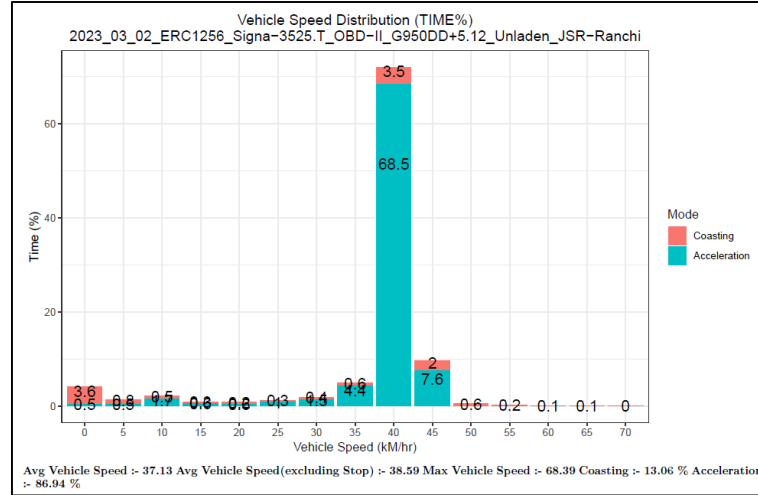


Vehicle Speed Distribution

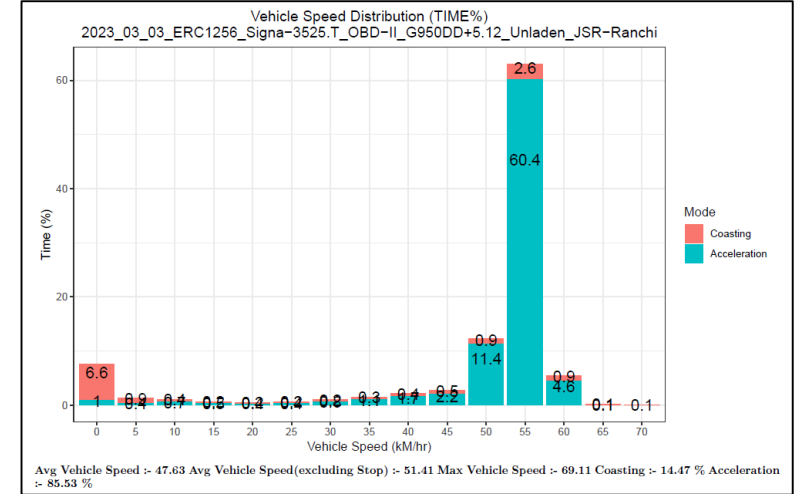
Unladen Trial

Signal 3525.T RDE Test Vehicle

02.03.23 40 kmph

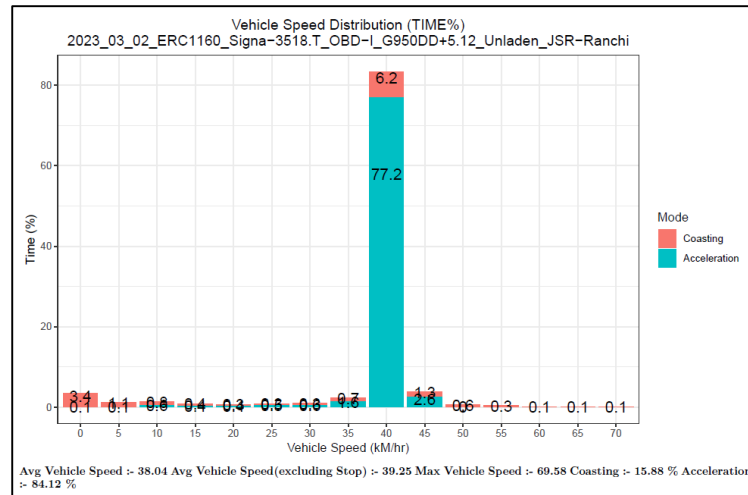


03.03.23 55 kmph

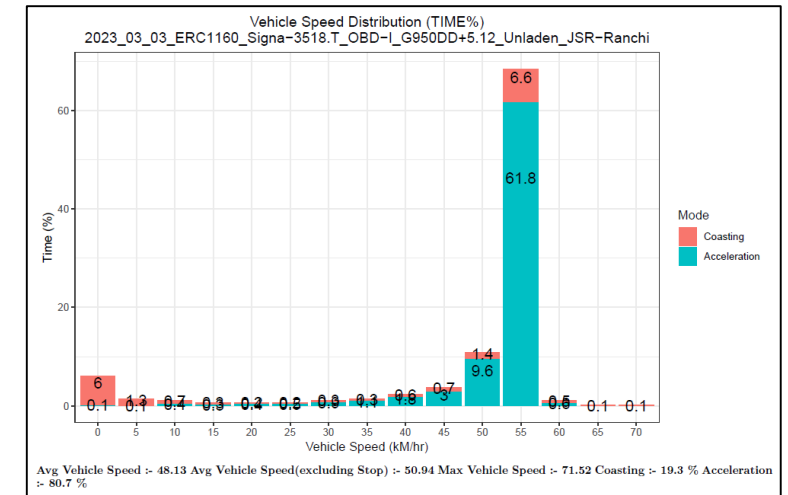


Signal 3518.T Base Vehicle

02.03.23 40 kmph



03.03.23 55 kmph

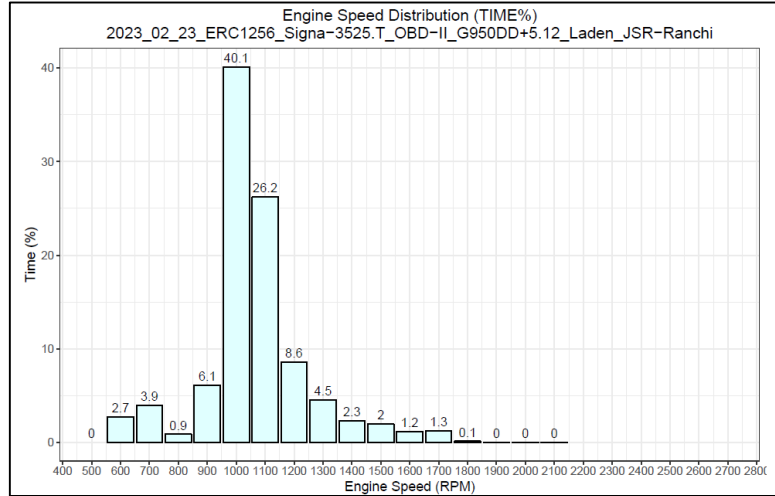


Engine Speed Distribution

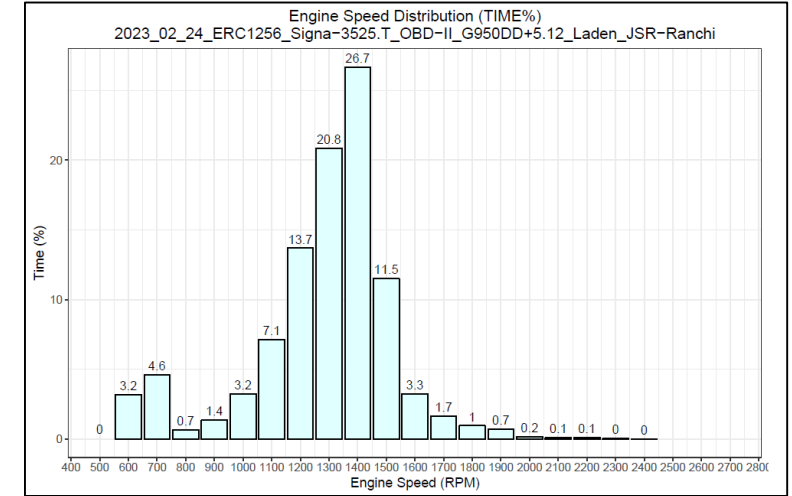
Laden Trial

Signa 3525.T RDE Test Vehicle

23.02.23 40 kmph

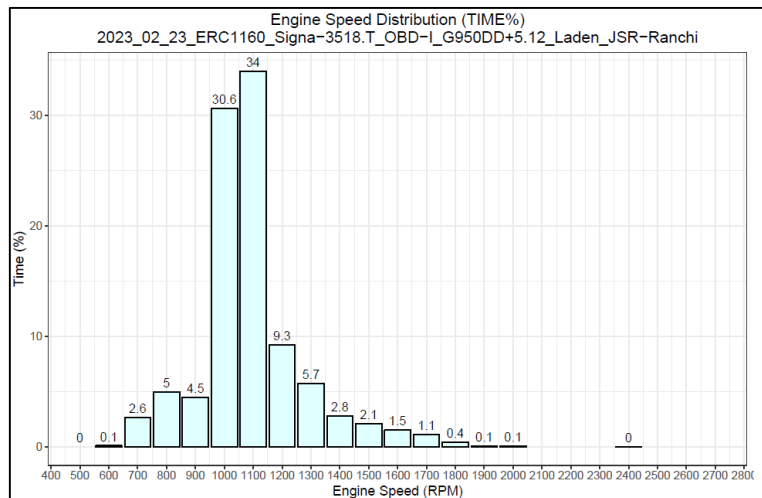


24.02.23 55 kmph

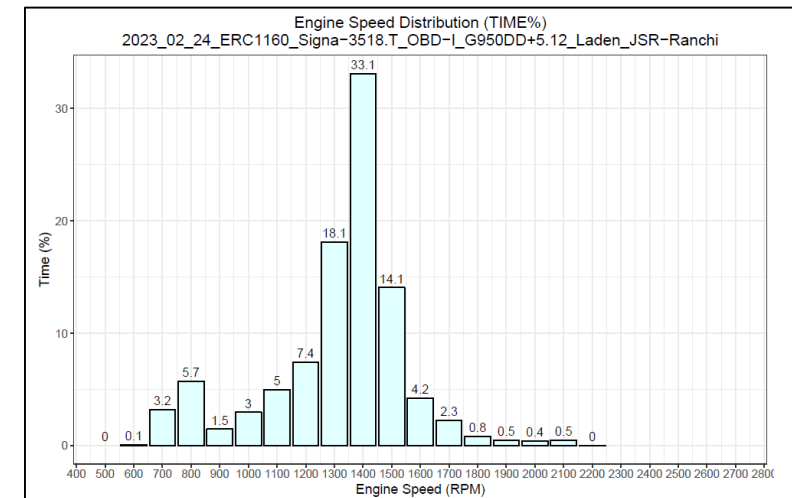


Signa 3518.T Base Vehicle

23.02.23 40 kmph



24.02.23 55 kmph

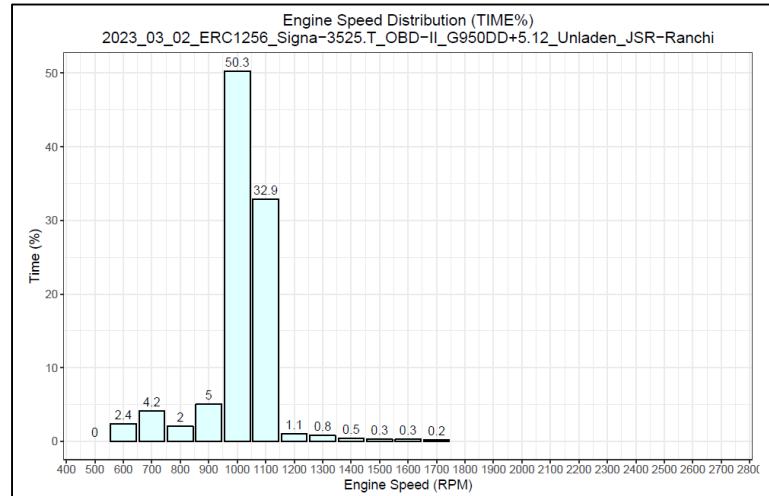


Engine Speed Distribution

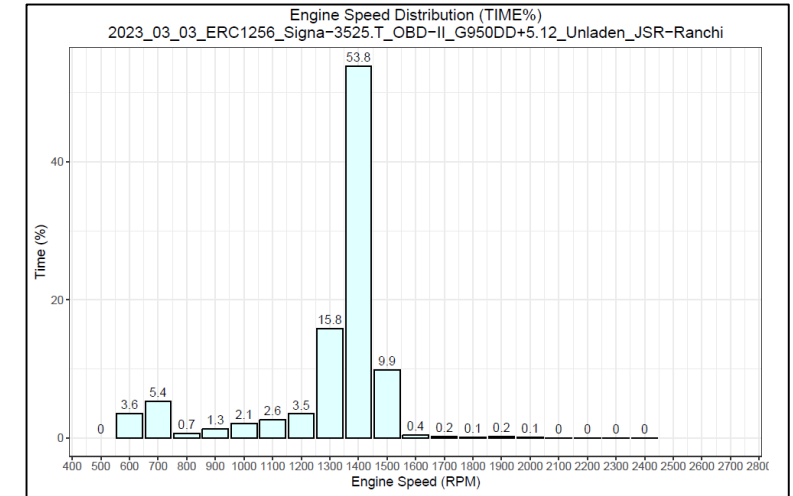
Unladen Trial

Signal 3525.T RDE Test Vehicle

02.03.23 40 kmph

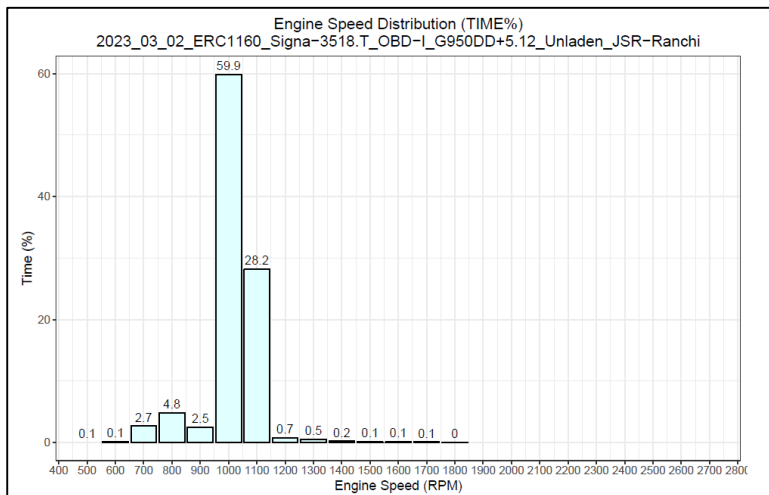


03.03.23 55 kmph

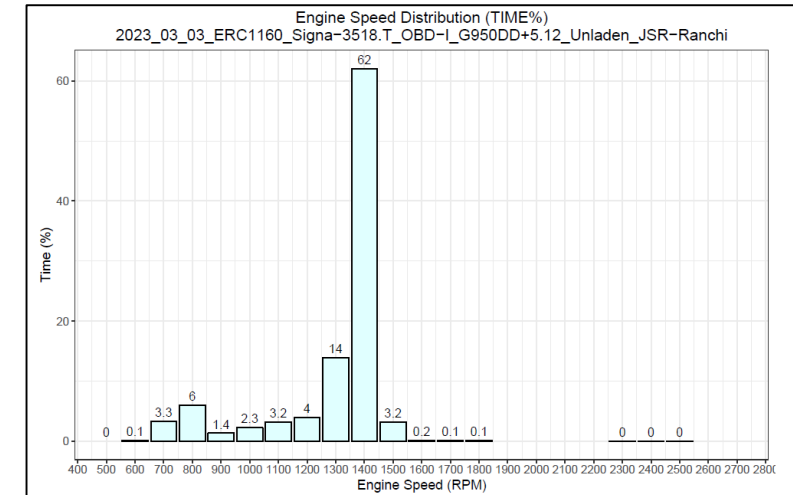


Signal 3518.T Base Vehicle

02.03.23 40 kmph



03.03.23 55 kmph

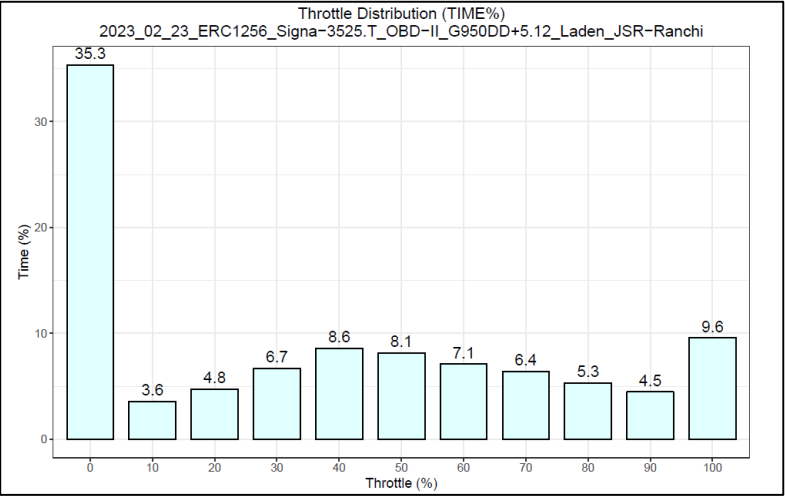


Throttle Distribution

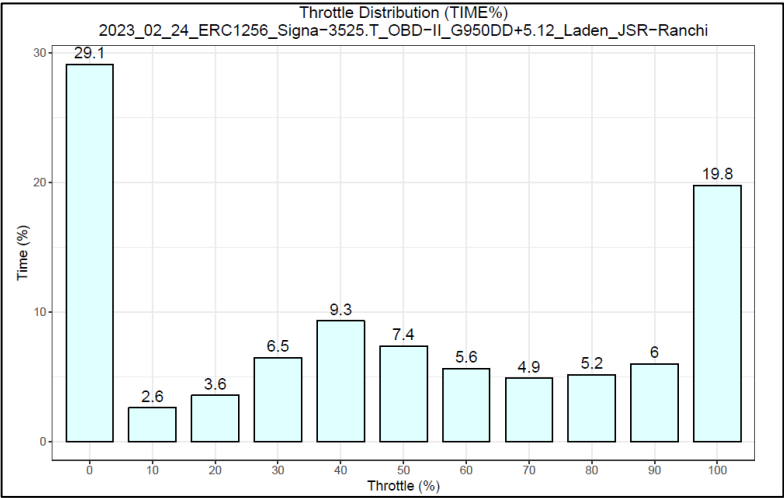
Laden Trial

Signal 3525.T RDE Test Vehicle

23.02.23 40 kmph

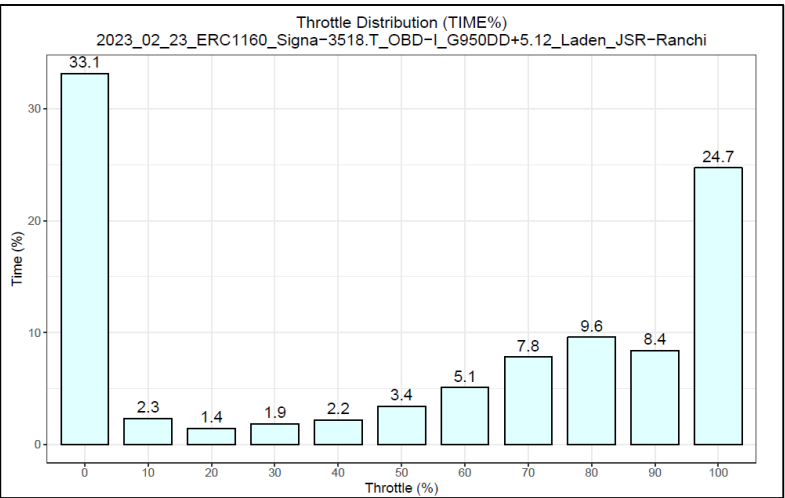


24.02.23 55 kmph

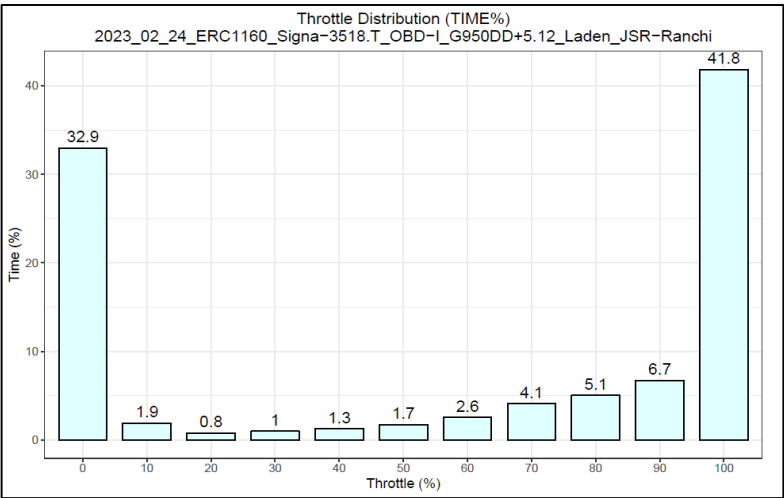


Signal 3518.T Base Vehicle

23.02.23 40 kmph



24.02.23 55 kmph



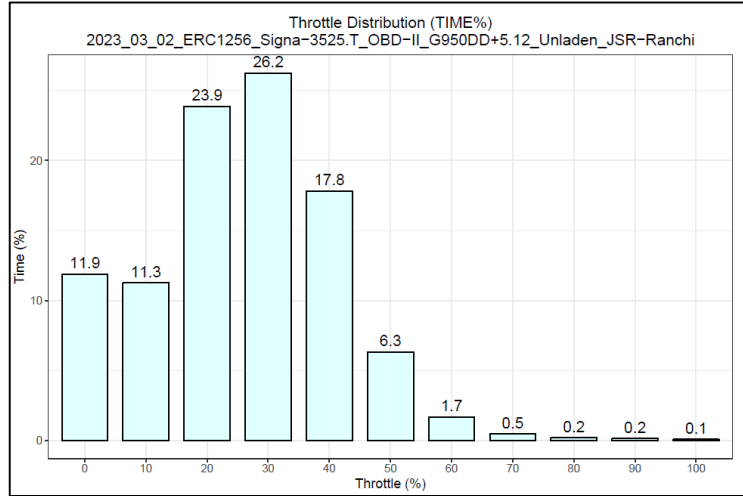
Throttle Distribution

Unladen Trial

Signal 3525.T RDE Test Vehicle

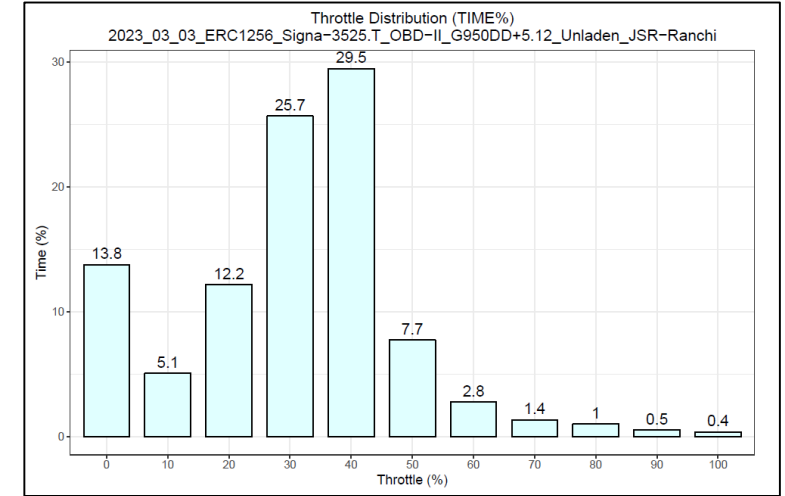
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40 kmph



03.03.23

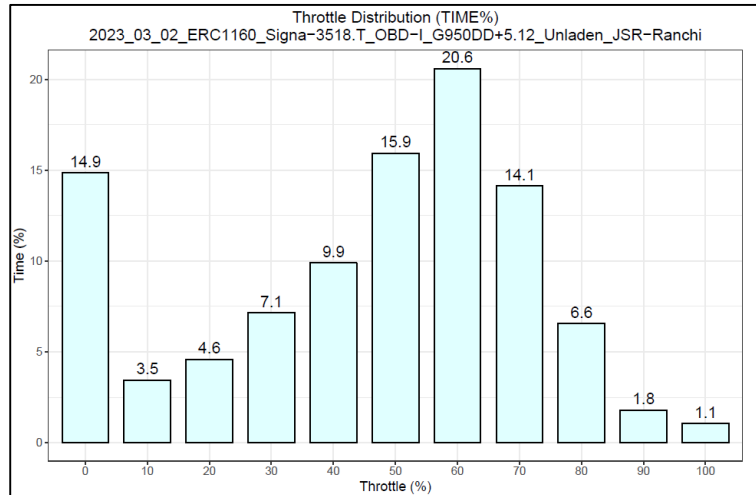
55 kmph



Signal 3518.T Base Vehicle

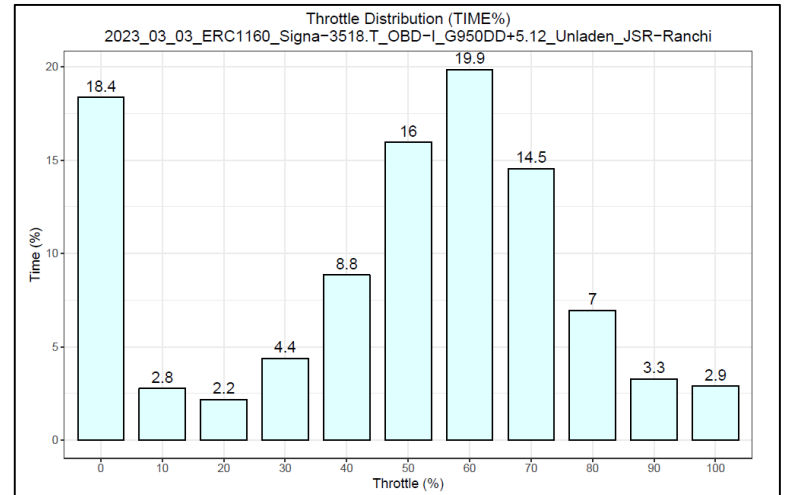
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40 kmph



03.03.23

55 kmph

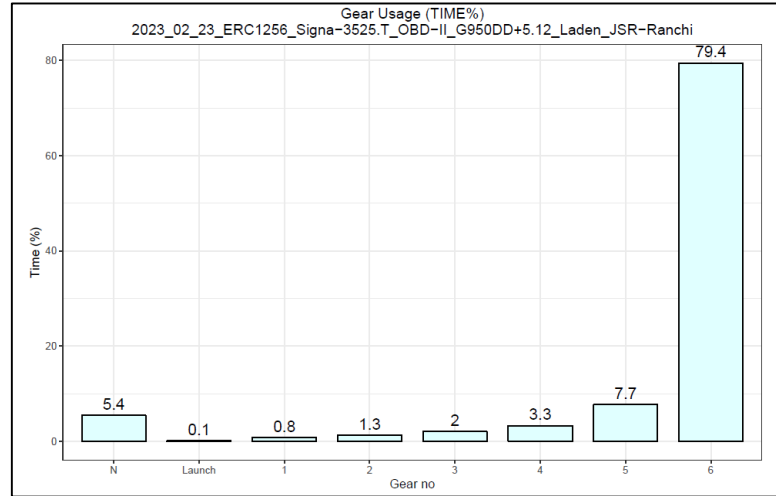


Gear Utilization

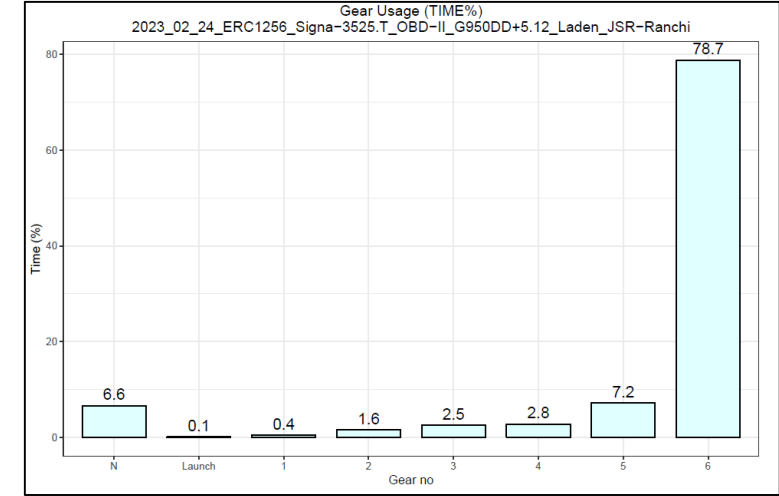
Laden Trial

Signa 3525.T RDE Test Vehicle

23.02.23 40 kmph

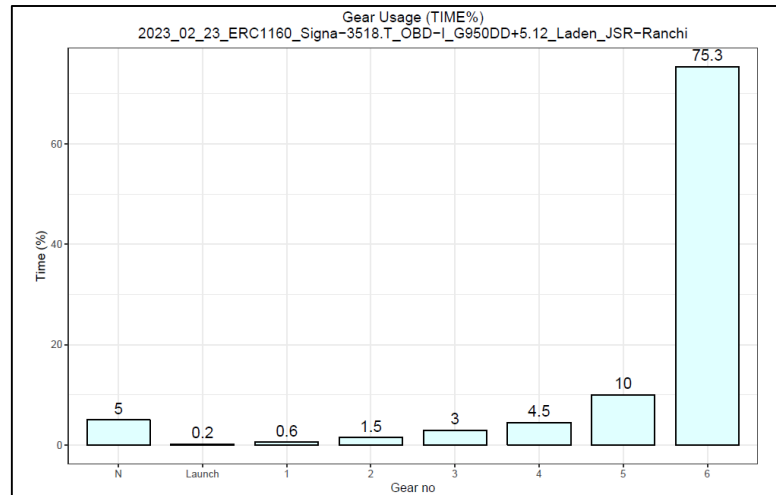


24.02.23 55 kmph

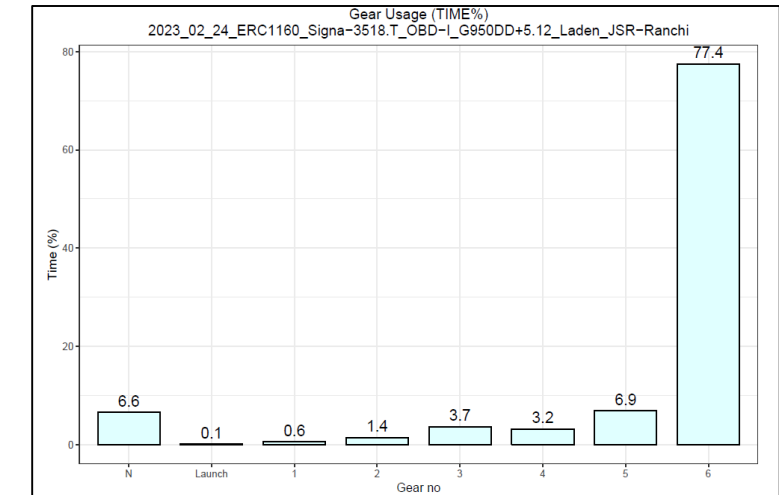


Signa 3518.T Base Vehicle

23.02.23 40 kmph



24.02.23 55 kmph

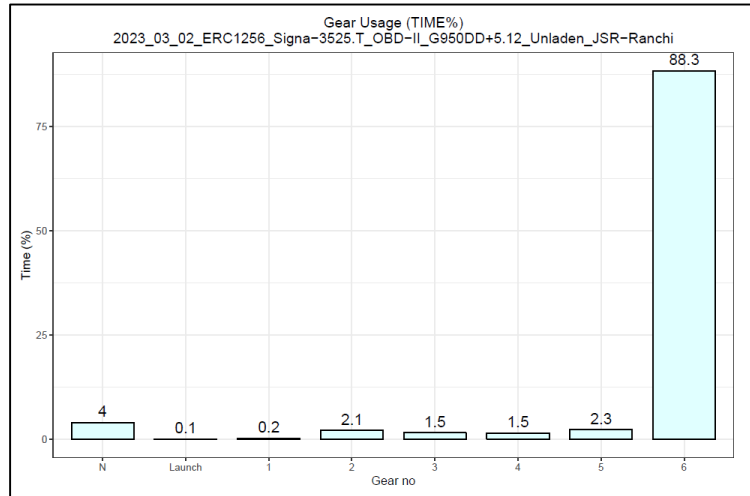


Gear Utilization

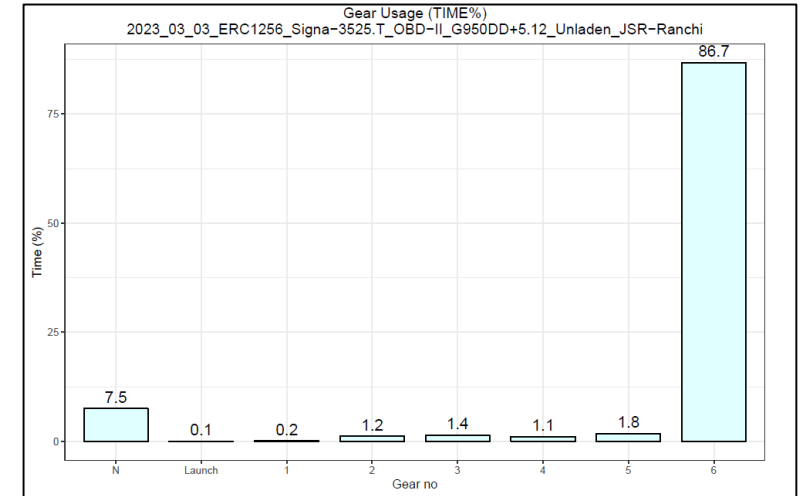
Unladen Trial

Signal 3525.T RDE Test Vehicle

02.03.23 40 kmph

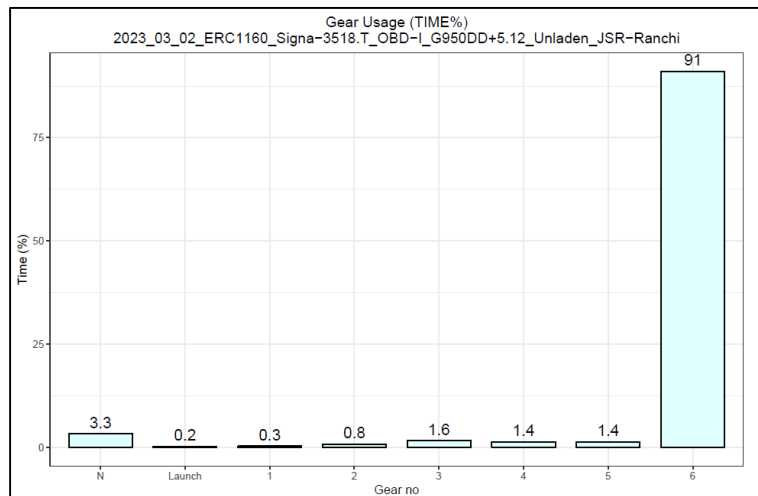


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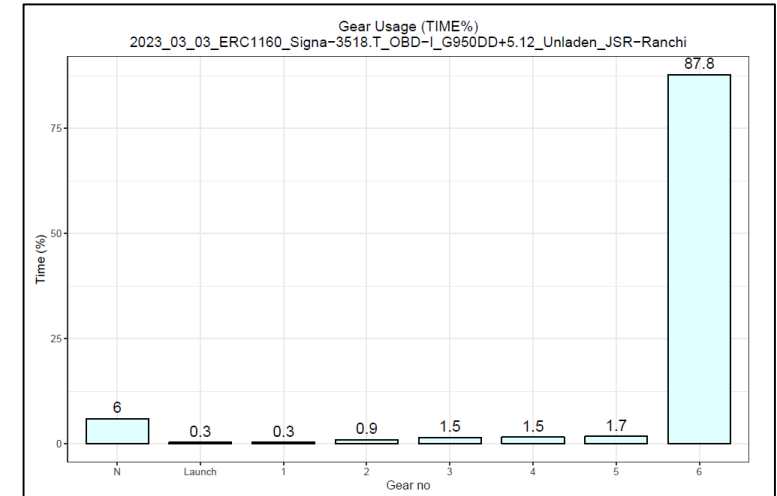


Signal 3518.T Base Vehicle

02.03.23 40 kmph



03.03.23 55 kmph



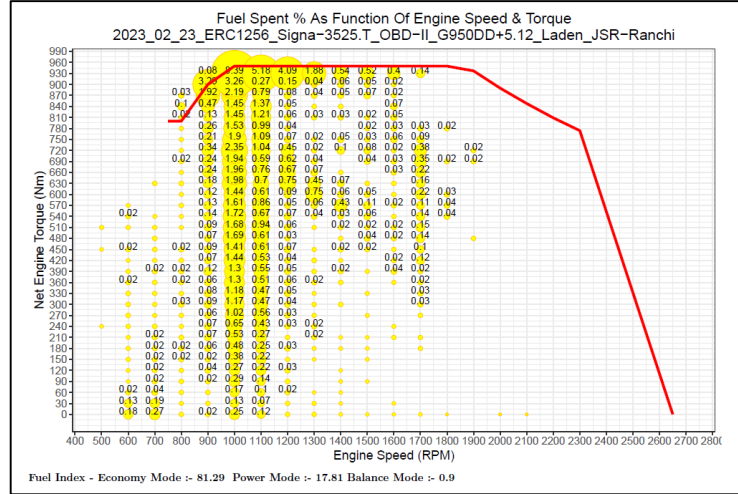
Fuel spent % as Engine speed & torque

Laden Trial

Signa 3525.T RDE Test Vehicle

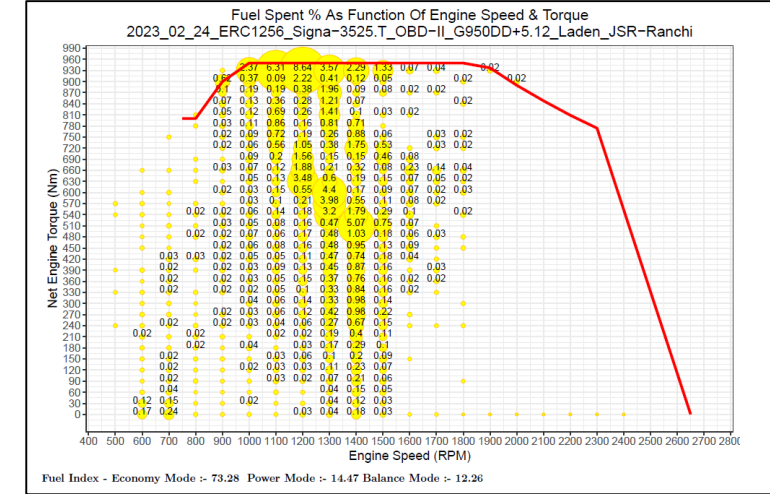
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40 kmph



24.02.23

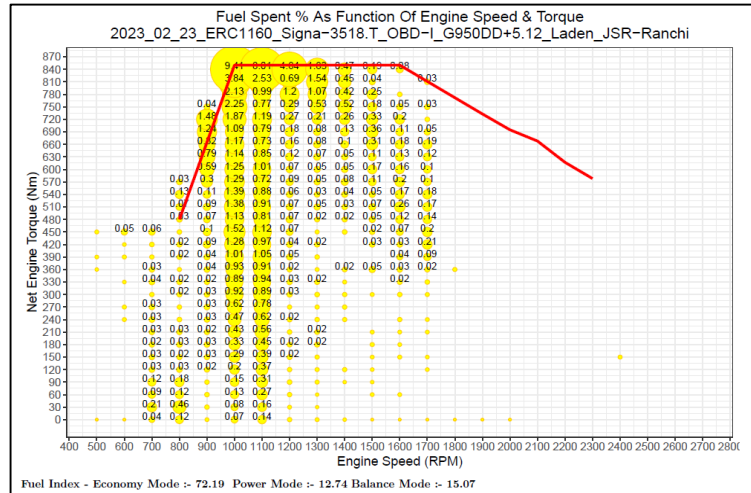
55 kmph



Signa 3518.T Base Vehicle

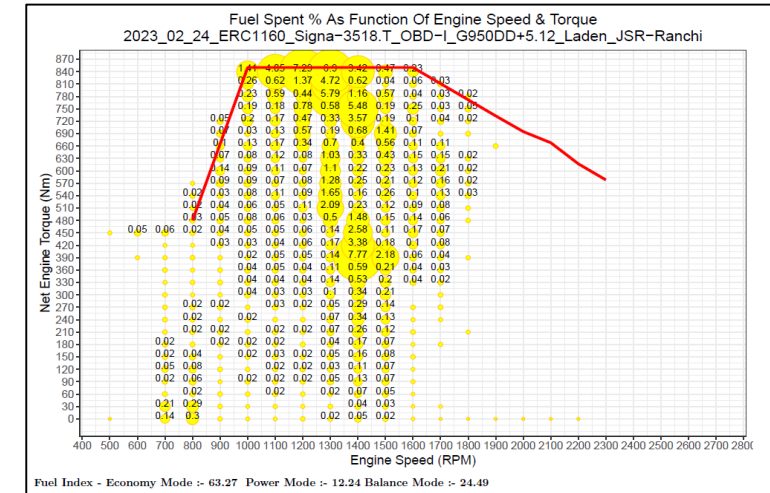
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40 kmph



24.02.23

55 kmph



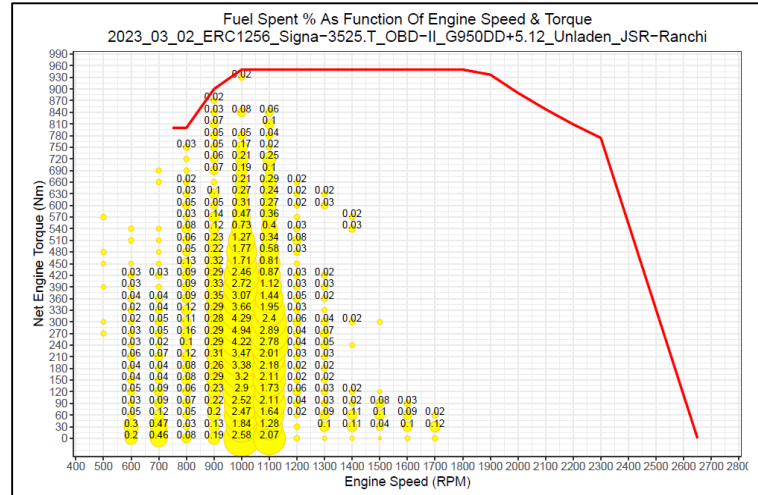
Fuel spent % as Engine speed & torque

Unladen Trial

Signa 3525.T RDE Test Vehicle

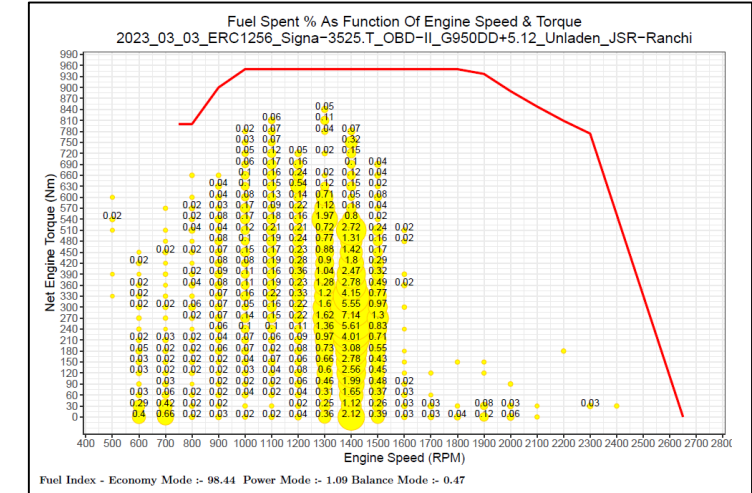
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40 kmph



03.03.23

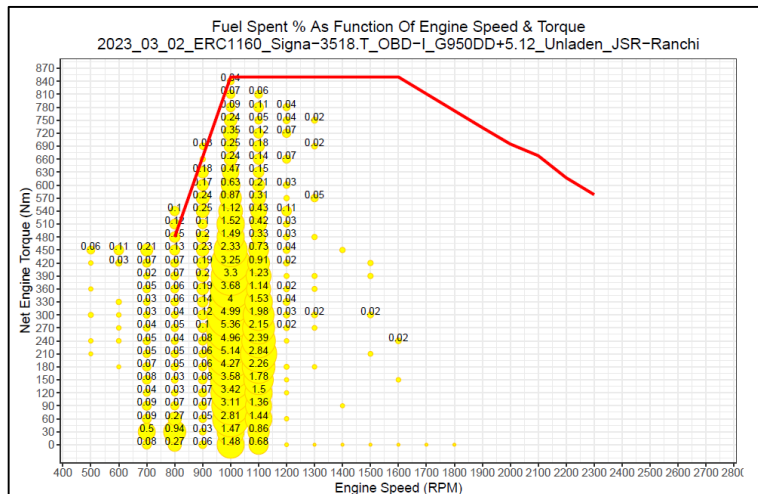
55 kmph



Signa 3518.T Base Vehicle

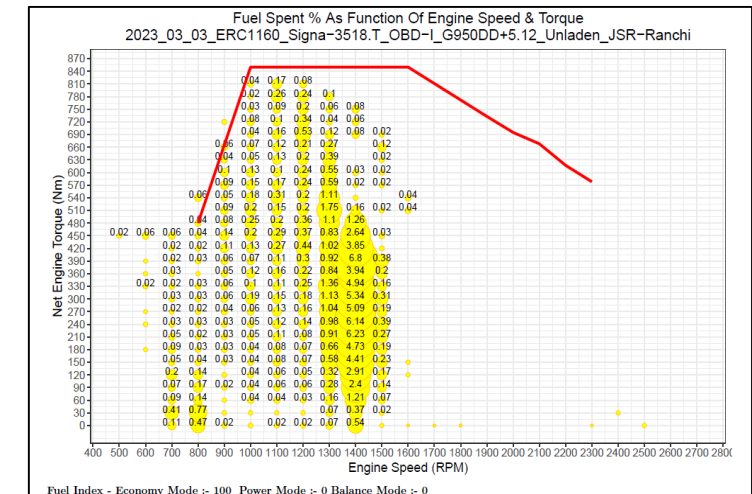
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40 kmph



03.03.23

55 kmph



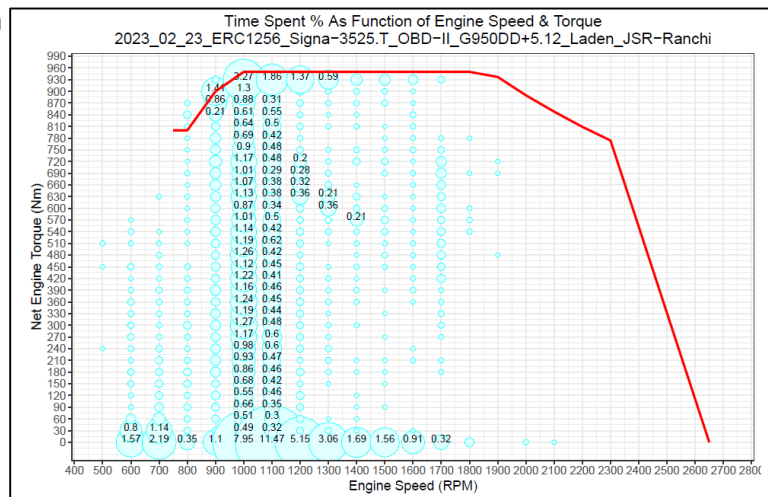
Time spent % as Engine speed & torque

Laden Trial

Signa 3525.T RDE Test Vehicle

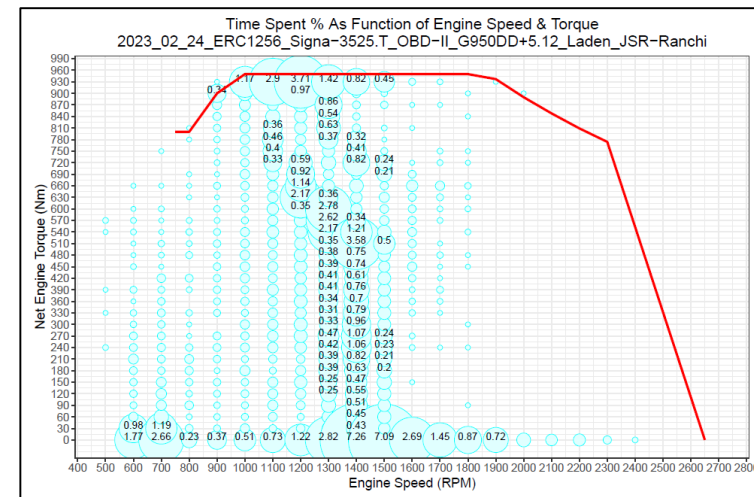
23.02.23

40 kmph



24.02.23

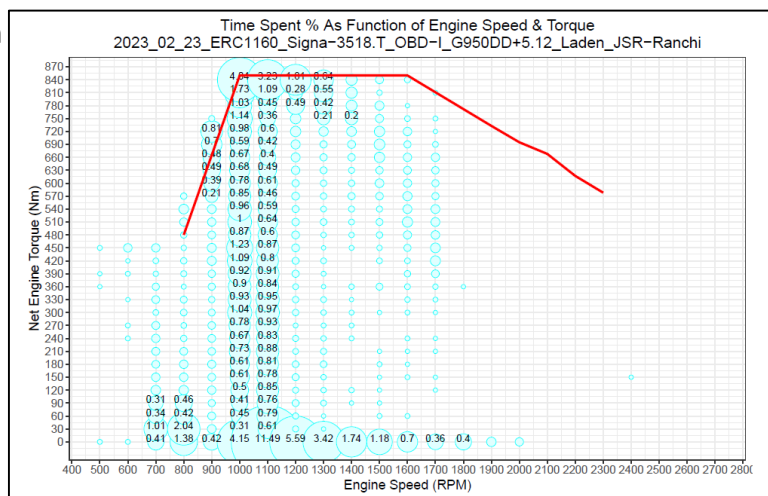
55 kmph



Signa 3518.T Base Vehicle

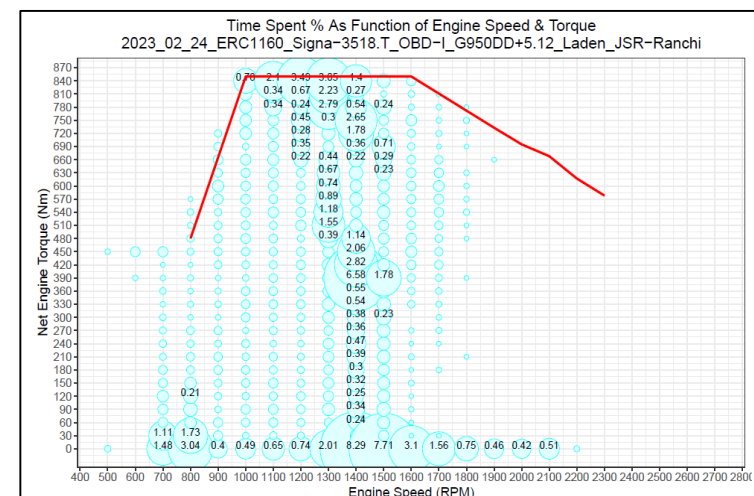
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40 kmph



24.02.23

55 kmph



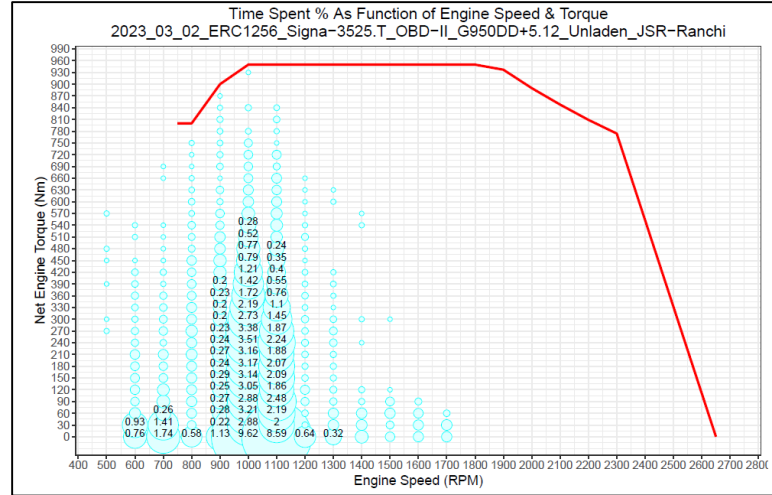
Time spent % as Engine speed & torque

Unladen Trial

Signa 3525.T RDE Test Vehicle

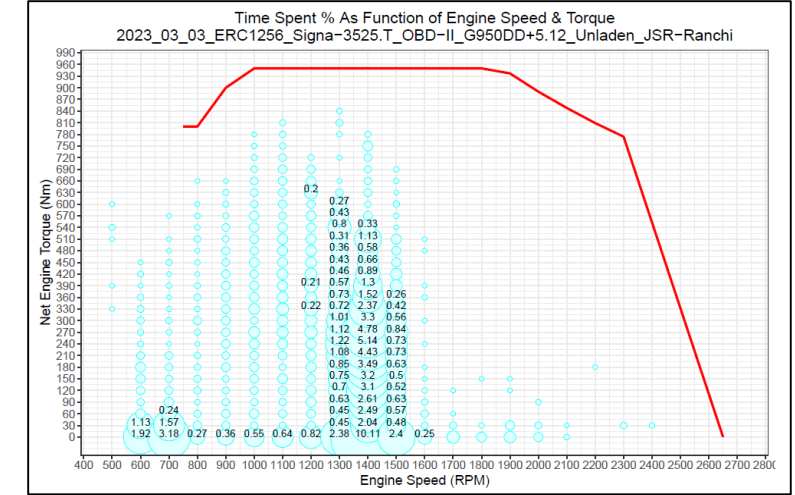
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40 kmph



03.03.23

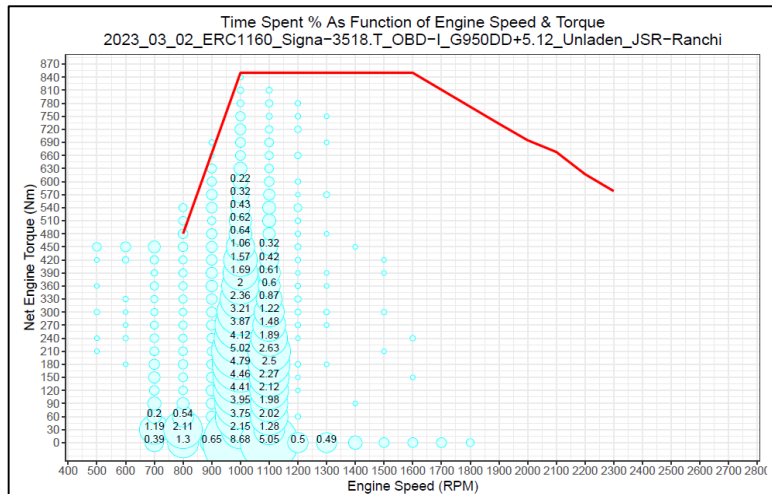
55 kmph



Signa 3518.T Base Vehicle

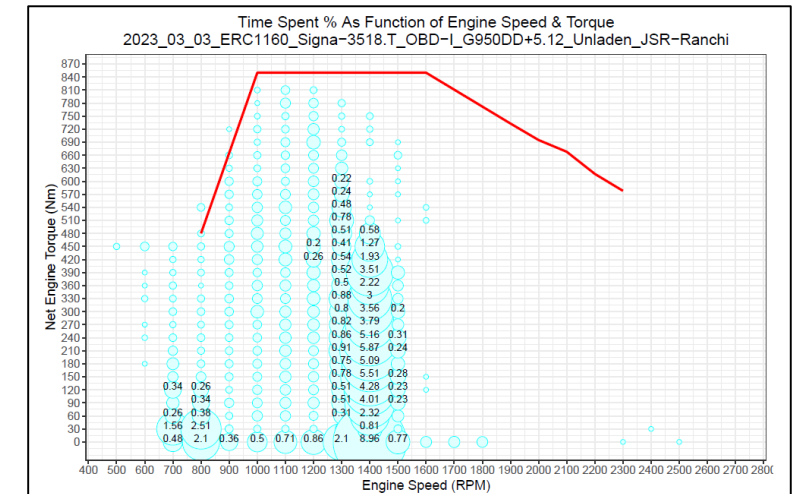
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40 kmph



03.03.23

55 kmph

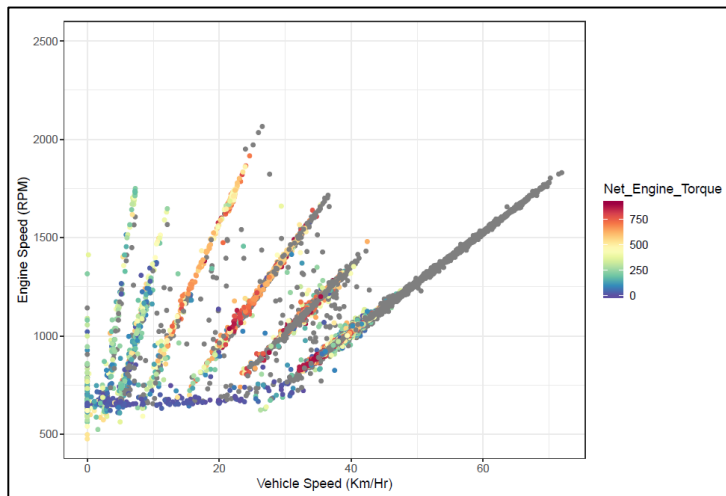


Transmission Shift Pattern

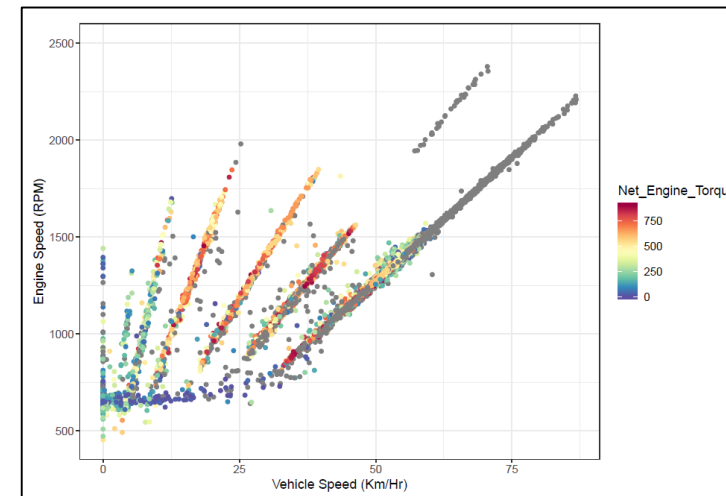
Laden Trial

Signa 3525.T RDE Test Vehicle

23.02.23 40 kmph

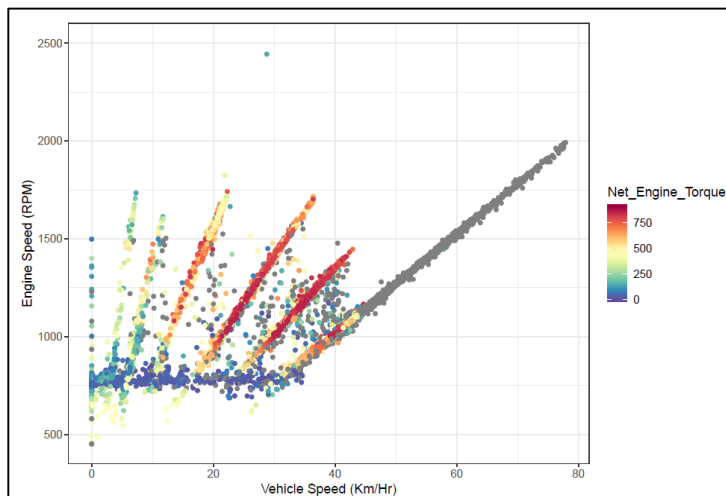


24.02.23 55 kmph

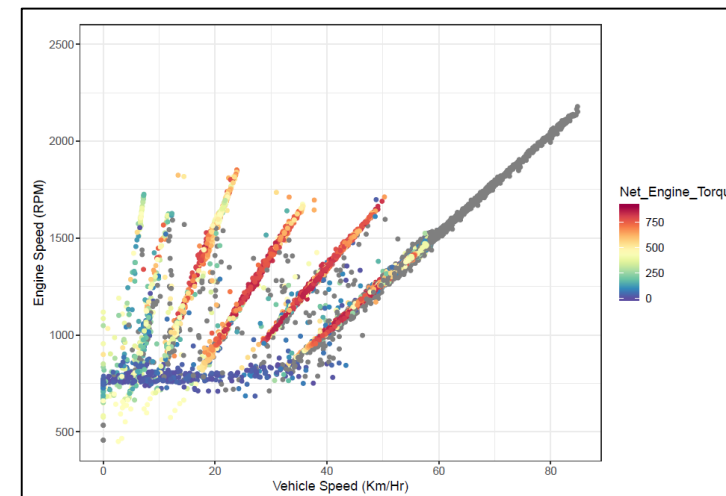


Signa 3518.T Base Vehicle

23.02.23 40 kmph



24.02.23 55 kmph

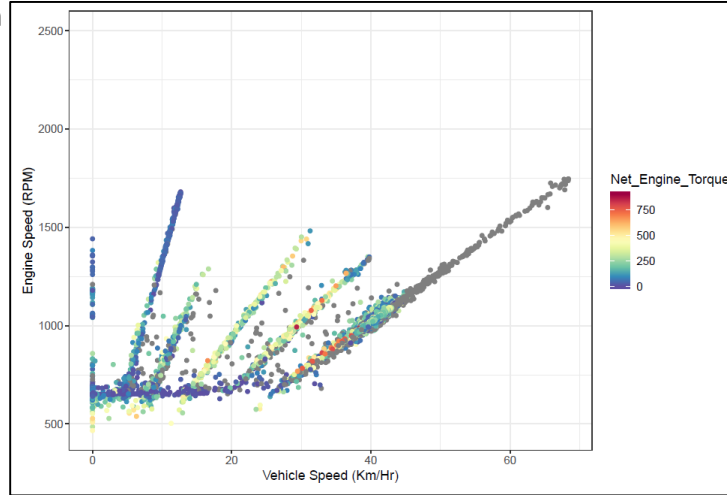


Transmission Shift Pattern

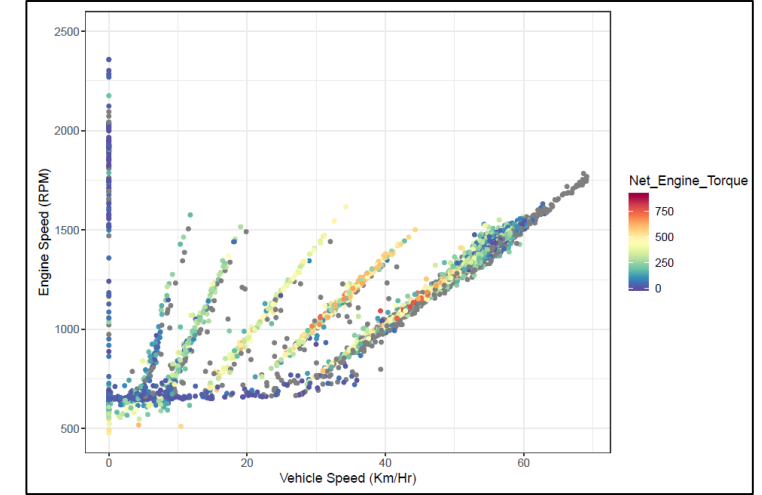
Unladen Trial

Signa 3525.T RDE Test Vehicle

02.03.23 40 kmph

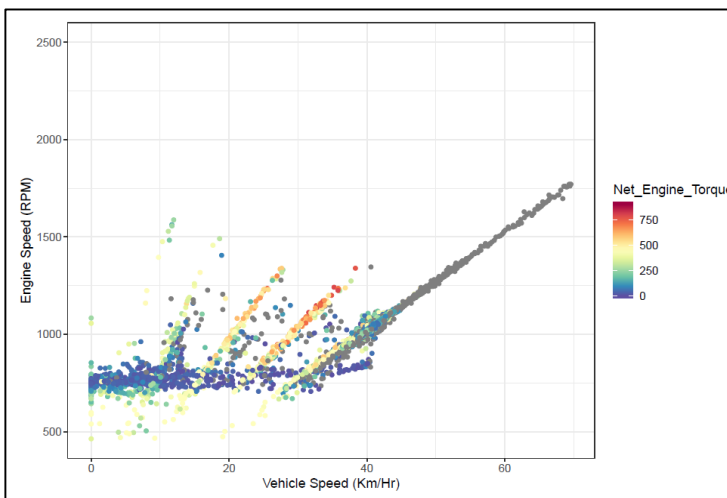


03.03.23 55 kmph

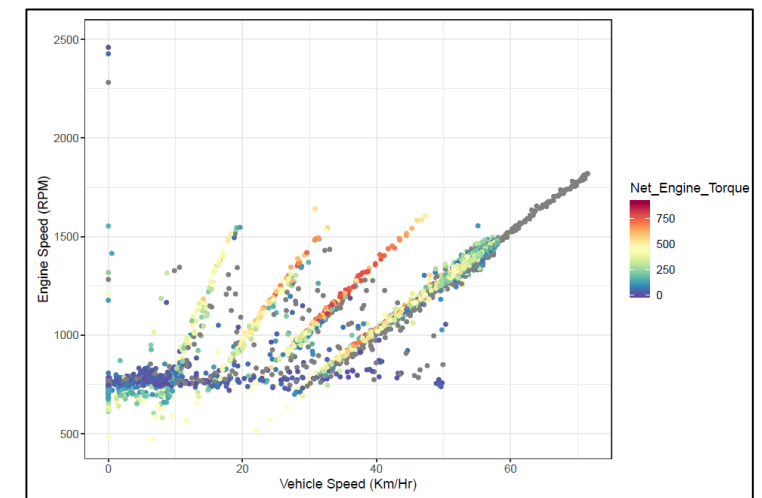


Signa 3518.T Base Vehicle

02.03.23 40 kmph



03.03.23 55 kmph



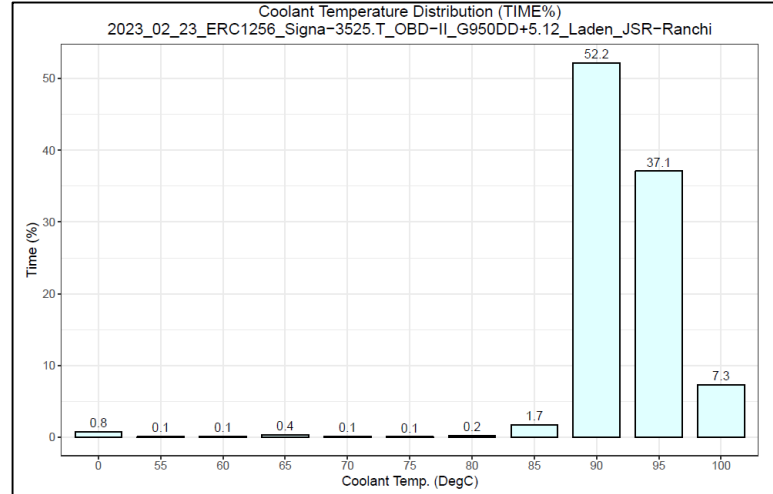
Coolant Temp Distribution

Laden Trial

Signal 3525.T RDE Test Vehicle

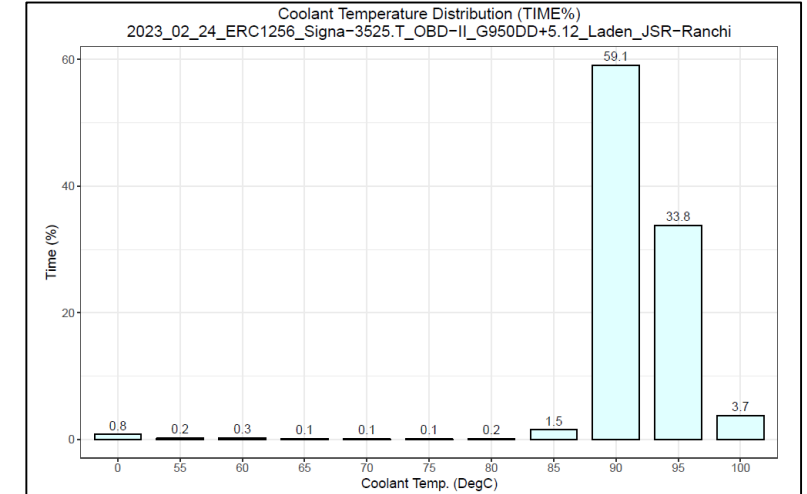
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40 kmph



24.02.23

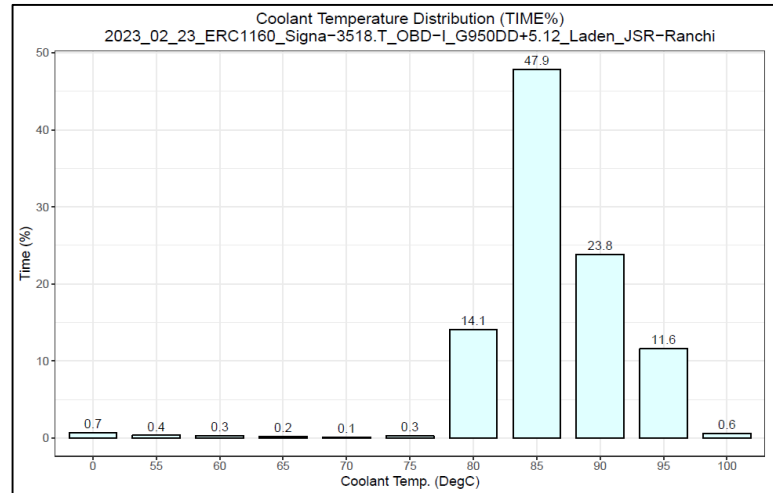
55 kmph



Signal 3518.T Base Vehicle

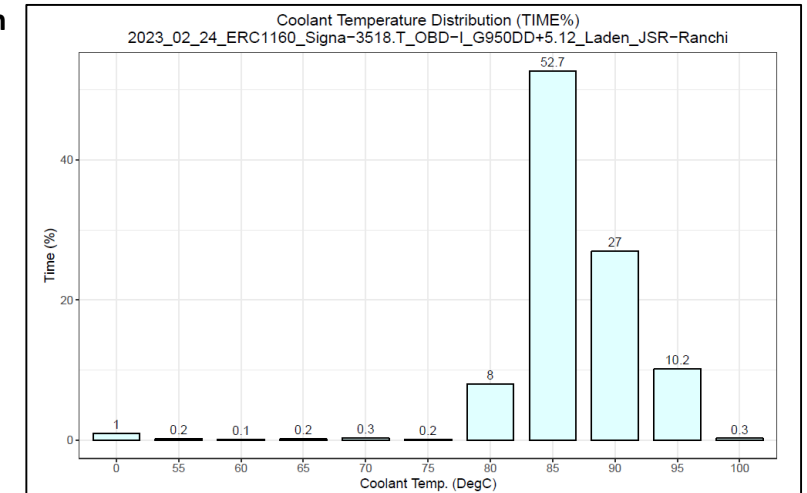
23.02.23

40 kmph



24.02.23

55 kmph

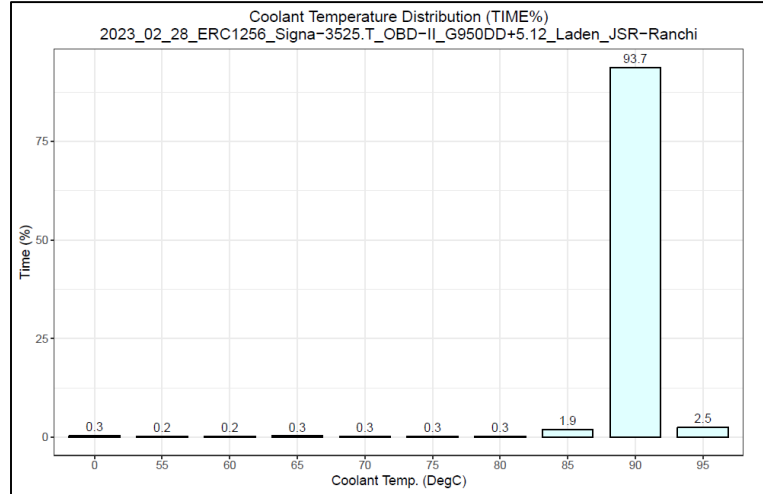


Coolant Temp Distribution

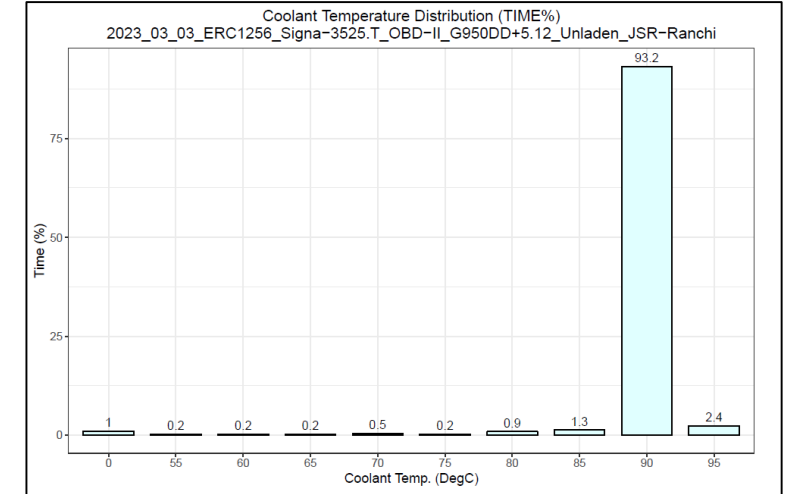
Unladen Trial

Signa 3525.T RDE Test Vehicle

28.02.23 40 kmph

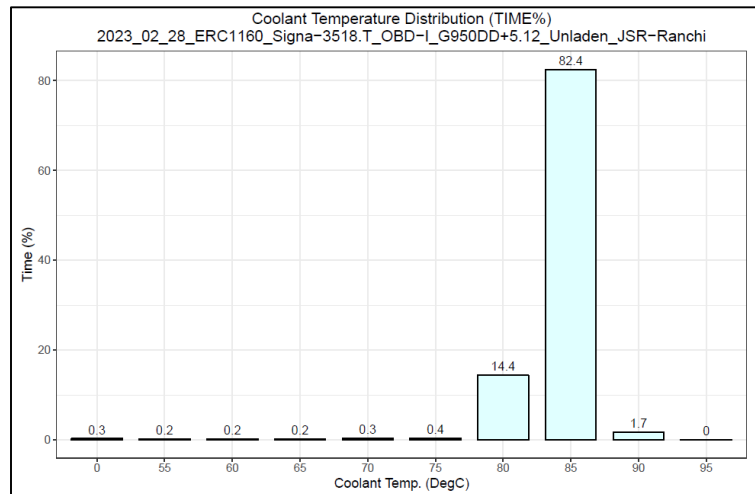


03.03.23 55 kmph

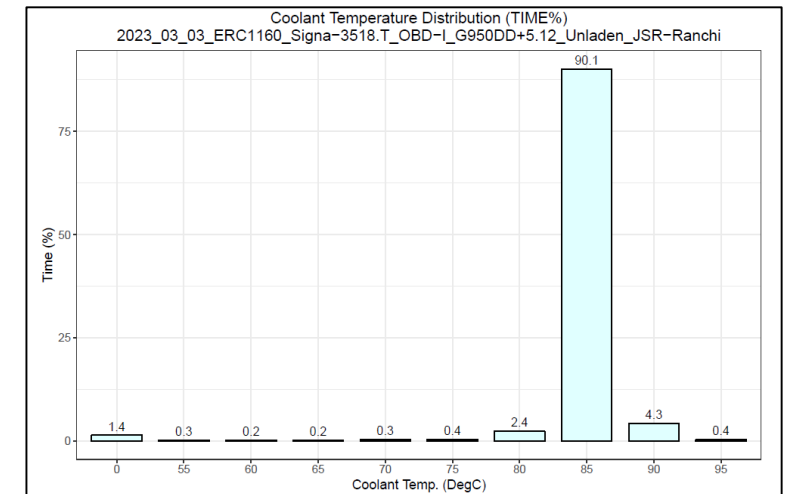


Signa 3518.T Base Vehicle

28.02.23 40 kmph



03.03.23 55 kmph



Fluid Economy(FLE) Status of 3525.T BSVI RDE Test vehicle w.r.t. 3518.T BSVI base vehicle :

1. Laden

- a) 40kmph :- Fluid Economy of 3525.T is **superior** than base vehicle by **5.97 %** .
- b) 55kmph :- Fluid Economy of 3525.T is **superior** than base vehicle by **5.18 %** .

2. Unladen

- a) 40kmph :- Fluid Economy of 3525.T is **superior** than base vehicle by **3.38 %**.
- b) 55kmph :- Fluid Economy of 3525.T is **superior** than base vehicle by **5.04 %**.

Overall Average Fluid Economy(FLE) Status of 3525.T BSVI RDE Test vehicle w.r.t. 3518.T BSVI base vehicle :

- a) **40kmph (Laden(80%) +Unladen(20%))**:- Fluid Economy of 3525.T is **superior** than base vehicle by **4.17 %** .
- b) **55kmph (Laden(80%) +Unladen(20%))**:- :- Fluid Economy of 3525.T is **superior** than base vehicle by **5.08 %** .

Thank You

 TATA MOTORS LTD. ERC,PUNE		PAT ATTRIBUTES SHEET - Fuel Economy										VE CVBU/FE/01							
												PAT Sheet No.: VE CVBU/ HCV/ SIGNA 3525.T 6.7L BSVI RDE/ Cx / FE/01							
		Depart ment Name: Vehicle Engineering		Project Name: SIGNA 3525.T 6.7L BSVI RDE_ Cx (G-950 DD FGR-9.36, 5.12 RAR, 295/90R 20 phase II low CRR Tyres, 3 Mode FE Switch) Base Vehicle: SIGNA 3518.T 5.6L BSVI OBD-1_G-950 DD, FGR-9.36, 5.12 RAR, 295/90R 20 Tyres_3 Mode FE Switch								Date: 23/05/2023							
Sr.No.	Aggregate and test parameter						Unit	Test Type	Classification @DR4	Ref.Standard / Work Guideline	Simulation	Base Vehicle	Target	Test Vehicle	Status (RYG)	Remarks	CoC Remark		
	Level 2	Level 3	Level 4	Level 5	Level 6	Level 7						SIGNA 3518.T BSVI OBD-1 (G 950 DD RAR 5.12 295/90R20)		SIGNA 3525.T BSVI OBD-II Cx (G 950 DD RAR 5.12 295/90R20)					
A	Fuel Economy Evaluation											Heading							
	LBSC - 1700,1700,35000 Kg VAM- 2.5 Kmph/s, 2.5 Kmph/s, 10 kmph, 40kmph GDP- 50 kmph, 50 kmph RSG - 80kmph													LBSC - 1700,1700,35000 Kg VAM- 1.5 Kmph/s, 1.5 Kmph/s, 10 kmph, 40kmph GDP- 45 kmph, 45 kmph RSG - 80kmph	--				
2.0	Fuel Economy At Rated Load											Heading							
2.7	Highway AC OFF								Heading										
2.7.1	Route : Jamshedpur-Ranchi							Standard	Specification	TST/VTWG/158						*Signa 3525.T OBDII Cx and Signa 3518.T OBDI was tested back to back condition On JSR-Ranchi route. *Signa 3525.T OBD-2 Cx Variant is with Non-THU bearing, so that FLE value is Extrapolated by 2% reduction of THU Bearing Contribution.			
2.7.9	Average Fuel Economy @40 kmph						Kmpl		Data Genration			3.57	--	3.78	--				
2.7.10	Average Fuel Economy @55 kmph						Kmpl		Data Genration			3.42	--	3.61	--				
2.7.11	Average Fluid Economy @40 kmph						Kmpl		Target			3.47		3.67	--				
	Average Fluid Economy @40 kmph (Extrapolated for Non-THU bearing)						Kmpl					≥ 3.47		3.60	G				
2.7.12	Average Fluid Economy @55 kmph						Kmpl		Target			3.30		3.47	--				
2.7.13	Average Fluid Economy @55 kmph (Extrapolated for Non-THU bearing)						Kmpl	Data Genration			≥ 3.30	3.40	G						
4.0	Fuel Economy At Unladen											Heading							
4.7	Highway AC OFF								Heading										
4.7.1	Route : Jamshedpur-Ranchi							Standard	Specification	TST/VTWG/158						*Signa 3525.T OBDII Cx and Signa 3518.T OBDI was tested back to back condition On JSR-Ranchi route.			
4.7.9	Average Fuel Economy @40 kmph						Kmpl		Data Genration			6.37	--	6.60	--				
4.7.10	Average Fuel Economy @55 kmph						Kmpl		Data Genration			5.79	--	6.07	--				
4.7.11	Average Fluid Economy @40 kmph						Kmpl		Target			≥ 6.19		6.40	G				
4.7.12	Average Fluid Economy @55 kmph						Kmpl		Target			≥ 5.62		5.90	G				
5.0	Overall Fuel Economy (80% Laden + 20% Unladen)											Heading							
5.1	Highway AC OFF								Heading										
5.1.1	Route : Jamshedpur-Ranchi							TST/VTWG/158	Specification							*Signa 3525.T OBDII Cx and Signa 3518.T OBDI was tested back to back condition On JSR-Ranchi route. *Signa 3525.T OBD-2 Cx Variant is with Non-THU bearing, so that FLE value is Extrapolated by 2% reduction of THU Bearing Contribution.			
5.1.9	Average Overall Fuel Economy @40 kmph						Kmpl		Data Genration		5.51	--	5.74	--					
5.1.10	Average Overall Fuel Economy @55 kmph						Kmpl		Data Genration		5.08	--	5.34	--					
5.1.11	Overall Average Fluid Economy @40 kmph						Kmpl		Target		5.35		5.57	--					
	Overall Average Fluid Economy @40 kmph (Extrapolated for Non-THU bearing)						Kmpl		Target		≥ 5.35		5.54	G					
5.1.12	Overall Average Fluid Economy @55 kmph						Kmpl		Target		4.92		5.17	--					
	Overall Average Fluid Economy @55 kmph (Extrapolated for Non-THU bearing)						Kmpl	Target			≥ 4.92	5.14	G						
Sign Off																		Index for Status	
Responsibility		Name		Signature				Off target, no resolution											
PAT Engineer		Palash Das/ Swapnil Jadhav		 				Marginally off target, resolution planned/low risk											
PAT Lead / Project Lead		Ajaykumar Jadhav / Sanjeev kumar		 				Met target, no concern											

Annexure-1

Level 1 Attribute:					PANEL Chart					VE CVBU/FE/01												
PAT- Fuel Economy					Specification:Max Power- 250 hp @ 2300 rpm Max Torque - 950 Nm 1000-1800 rpm Driveline : G-950 DD 5.12 RAR 295/90R 20 Tyres 3 Mode FE Switch Torque Factor: E-100% ; B-100% EOL Parameters : LBSC: 1700,1700 RPM VAM: 1.5,1.5 kmph/s @ 10,40 kmph GDP: 45,45 kmph RSG: 80 kmph					Panel Chart No: VE CVBU/ HCV/ SIGNA 3525.T 6.7L Cx BSVI RDE/ Cx / FE/01												
Programme / MY : SIGNA 3525.T Cx 6.7L BSVI RDE					Gateway : DR3		Owner: Ajaykumar Jadhav			Project Name: SIGNA 3525.T 6.7L BSVI RDE(Cx Version)			Date: 23/05/2023									
Level 1 Summary:					Statement Of Intent (SOI) or DNA Description:																	
<table><tr><td rowspan="2">PALS Position</td><td colspan="2">Status</td></tr><tr><td>Current/ Gateway</td><td>Job#1 Predicted</td></tr><tr><td>Level 1:</td><td>L</td><td>L</td></tr></table>					PALS Position	Status		Current/ Gateway	Job#1 Predicted	Level 1:	L	L	Fuel Economy: Haulage Application									
PALS Position	Status																					
	Current/ Gateway	Job#1 Predicted																				
Level 1:	L	L																				
Attribute Target Summary:										Remarks												
Deriv.						PALS	Metric	Target	Status		Base Vehicle											
	Level 2	Level 3	Level 4	Level 5	Level 6				Current/Gateway SIGNA 3525.T Cx BSVI RDE	Job#1 Predicted	Base Signa 3518.T BSVI											
	Specification								LBSC - 1700,1700,35000 Kg VAM- 1.5 Kmph/s, 1.5 Kmph/s, 10 kmph, 40kmph GDP- 45 kmph, 45 kmph RSG - 80kmph		LBSC - 1700,1700,35000 Kg VAM- 2.5 Kmph/s, 2.5 Kmph/s, 10 kmph, 40kmph GDP- 50 kmph, 50 kmph RSG - 80kmph											
	Vehicle GVW (Payload)						kg		35000		35000											
	Power (Max) @ rpm						HP		250 hp @ 2300 rpm		180 hp @ 2300 rpm											
	Power to Weight ratio						HP/Ton		7.14		5.14											
	Torque (Max) @ rpm						Nm		950 Nm @ 1000-1800 rpm,		850 Nm @ 1000-1600 rpm											
	Torque to Weight ratio						Nm/Ton		27.14		24.29											
	Tyres size						--		Bridgestone 295/90R20		Apollo 295/90R20											
	Gear Box Ratios						--		G950 DD, FGR -9.36		G950 DD, FGR -9.36											
	Final Drive Ratio						--		5.12		5.12											
	Fluid Economy at Rated Load					L			L													
	Highway AC OFF						--															
	Route : Jamshedpur-Ranchi						--															
	Avg. Fluid Economy @40 kmph						kmpl		3.67		3.47	*Signa 3525.T OBd1 Cx and Signa 3518.T OBd1 was tested back to back condition on JSR-Ranchi route.										
	Avg. Fluid Economy @40 kmph (Extrapolated for Non-THU bearing)							≥ 3.47	3.60													
	Avg. Fluid Economy @55 kmph						kmpl		3.47		3.30	*Signa 3525.T OBd-2 Cx Variant is with Non-THU bearing, so that FLE value is Extrapolated by 2% reduction of THU Bearing Contribution.										
	Avg. Fluid Economy @55 kmph (Extrapolated for Non-THU bearing)							≥ 3.30	3.40													
	Fluid Economy At Unladen					L			L													
	Highway AC OFF						--															
	Route : Jamshedpur-Ranchi						--															
	Avg. Fluid Economy @40 kmph						kmpl	≥ 6.19	6.40		6.19	*Signa 3525.T OBd1 Cx and Signa 3518.T OBd1 was tested back to back condition on JSR-Ranchi route.										
	Avg. Fluid Economy @55 kmph						kmpl	≥ 5.62	5.90		5.62											
	Overall Fuel Economy (80% Laden + 20% Unladen)					L			L													
	Highway AC OFF						--															
	Route : Jamshedpur-Ranchi						--															
	Overall Avg. Fluid Economy @40 kmph						kmpl		5.57		5.35	*Signa 3525.T OBd1 Cx and Signa 3518.T OBd1 was tested back to back condition on JSR-Ranchi route.										
	Overall Avg. Fluid Economy @40 kmph(Extrapolated for Non-THU bearing)						kmpl	≥ 5.35	5.46													
	Overall Avg. Fluid Economy @55 kmph						kmpl		5.17		4.92	*Signa 3525.T OBd-2 Cx Variant is with Non-THU bearing, so that FLE value is Extrapolated by 2% reduction of THU Bearing Contribution.										
	Overall Avg. Fluid Economy @55 kmph(Extrapolated for Non-THU bearing)						kmpl	≥ 4.92	5.07													
Executive Summary: 1. Test vehicle SIGNA 3525.T Cx BSVI OBd-2 tested with : 250 HP Engine calibration with 95% Cal Maturity + THU Bearing + Low viscous 15W50 engine oil + Low viscous 80W90 rear axle oil + Low idle 650 RPM + Tuned VAM + TTT (fan table & new thermostat of 89°C) + SCRTM Strategy change for low load operation + Bridgestone LCRR tyres + 24" Fan + Modified USFE table + Modified Calibration (POC-1 CAL) 2. Signa 3525.T OBd-2 Cx Variant is with Non-THU bearing, so that FLE value is Extrapolated by 2% reduction of THU Bearing Contribution. 3. Target has been taken at-par (As per Cx approach) wrt to base SIGNA 3518.T OBd-1 BSVI vehicle. 4. SIGNA 3525.T Cx RDE BSVI is meeting the PALS target of "Leadership".																						
Tested by:																						
Verified and Approved by:																						
 Palash Das / Swapnil Jadhav PAT ENGINEER  Ajaykumar Jadhav / Sanjeev Kumar PAT LEAD / PROJECT LEAD																						

TEST REQUISITION

Ref.	NA	Date:	30/03/2023
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To,

ERC Testing

Kindly arrange to take up for testing of the following component.

1	Part Description	TATA SIGNA 3525.T BSVI				
2	Part Number	51685253R,				
3	Drawing number	516800000006,	Mod status	A,2,	Mod Date	25/04/2023
4	Vehicle Model / Project	TATA SIGNA 3525.T RDe BSVI				
5	WBS Number	P.22.EA.501 -RDE Truck 5.6 Ltr 3123.T / 3523.T				
6	Supplier	NA				
7	Number of samples					
8	Inspection report	Enclosed	☒ Yes	☐ No	If No commitment date:	
9	Material report	Enclosed	☐ Yes	☒ No	If No commitment date:	
10	Suppliers test report	Enclosed	☒ Yes	☐ No	If No commitment date:	
11	Reason for testing	First Source Development	☐	Weight Reduction	☐	
		Alternate source development	☐	Cost Reduction	☐	
		Data generation Test	☐	Regulatory test	☐	
		Any other (pl. specify)	☐	New Development	☐	
12	What testing required?	DYNAMIC				
13	Remarks					
14	Peripheral Parts Availability	YES				

Authorized Signature:	
Name:	
Dept:	

Contact Person:	Prithviraj Ray
Phone Number	

For Testing Dept Use only	Test Item Number and Date(stamp)	Test Engineer

TATA MOTORS LIMITED ERC PUNE	TEST PLAN		DATE: 30/04/2023			
	VEHICLE TESTING DEPARTMENT					
Description: Signa 3525.T 5.6L RDE BSVI Cx			Test Item No: 41816			
			New	✓	Moderate	✗
			Major	✗	Minor	✗

Part No: 51685253R,		Modification No: A,2,	Enclosures: Back to back FE Trials of Signa 3525.T 5.6L RDE BSVI Cx Vehicle
Drawing No: 5168000000006,			
Make: NA		WBS Number: P.22.EA.501 -RDE Truck 5.6 Ltr 3123.T / 3523.T	Copies to :
Vehicle Type: TATA SIGNA 3525.T RDe BSVI			
Test Requested by:	VEHICLE-I		Equipments:
Test Requisition Reference:	Testing-Outdoor/15045		
Reason for Test/s:	Component & System Level Performance Tests		
Regulatory Requirement:	a) Component:		
	b) Vehicle:		

Sr. No.	Test description	Test specification	Acceptance criteria	No. of Samples	Test Responsibility	Proto Phase (Mule/ Alpha/ Beta/PP)	Remarks
1	B2B FE Trial on Signa 3525.T RDE BSVI FE trial on Signa 3525.T RDE BSVI Vehicle	TST/VT/WG/158	As per PALS	1	Palash Das	Beta	NIL

REMARKS: FE Trials	INTERNAL REF:	
TEST TO FAILURE: No		
MEOST: No		
RELIABILITY TESTING: No		
BENCHMARKING TESTING: No		
DFMEA/FMEA DONE: Yes		
PREPARED BY	CHECKED BY	APPROVED BY

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Annexure 6

VC No.					VC : 51685156R	VC : 51685256R
Vehicle Model					Signa 3525.T.AC	Signa 3525.T.Blower
Parameter	VEPS Name	DEFAULT VALUE *	MIN/MAX *	UNITS	Signa 3525.T Atulya (G-950 DD [5.12 RAR [295/90R20])	Signa 3525.T Atulya (G-950 DD [5.12 RAR [295/90R20])
Fuel Economy						
Gear Down Protection (GDP)						
Gear Down Protection Enable	20061	N		Y = Enable N = Disable	Y	Y
Gear Down Protection Heavy Load Vehicle Speed	11440	28	Min: 16 Max: (10000)	mph	45 kmph (27.96 mph)	45 kmph (27.96 mph)
Gear Down Protection Light Load Vehicle Speed	11430	28	Min: 19 Max: 56	mph	45 kmph (27.96 mph)	45 kmph (27.96 mph)
Idle Speed Control						
Adjustable Low Idle Speed Enable (Read Only)	50512	Y		49	Y	Y
Low Idle Speed	10030	750	Min: 600 Max: 800	rpm	650	650
Idle Shutdown						
Idle Shutdown Enable	20051	N		Y = Enable N = Disable	Y	Y
Idle Shutdown Timer	10170	60.0	Min: 0.5 Max: 1440.0	Minutes	5	5
Idle Shutdown in PTO Enable	20071	N		Y = Enable N = Disable	N	N
Idle Shutdown Manual Override Enable	20081	N		Y = Enable N = Disable	Y	Y
Idle Shutdown service brake switch interaction	50057	N		Y = Enable N = Disable	Y	Y
Load-Based Speed Control						
Load Based Speed Control Enable	50016	N		Y = Enable N = Disable	Y	Y
Load Based Speed Control Low Engine Speed Breakpoint	50027	1800	Min: 1200 Max: 2100	rpm	1800	1800
Load Based Speed Control High Engine Speed Breakpoint	50028	1800	Min: 1500 Max: 2500	rpm	1800	1800
Load Based Speed Control Vehicle Mass Threshold Enable (Read Only)	50446	Y		Y = Enable N = Disable	Y	Y
Load Based Speed Control Vehicle Mass Threshold	50447	68343	Min: 22046 Max: 220460	lb	35000 kg (77161.79 lbs)	35000 kg (77161.79 lbs)
Road Speed Governor						
Road Speed Governor Enable (Read Only)	99001	Y		Y = Enable N = Disable	Y	Y
Maximum Accelerator Vehicle Speed	10000	56	Min: 12 Max: (13590)	mph	80 kmph (49.709 mph)	80 kmph (49.709 mph)
User Adjustable Fuel Economy						
User Adjustable Fuel Economy Enable	50437	N		Y = Enable N = Disable	Y	Y
UAFE Max Road Speed Enable	50475	Y		Y = Enable N = Disable	Y	Y
UAFE Max Road Speed Limit	50476	36	Min: 12 Max: 56	mph	79 kmph (49.09 mph)	79 kmph (49.09 mph)
UAFE Torque Limit Enable	50486	N		Y = Enable N = Disable	Y	Y
UAFE Mid Torque Limit Adjust Factor	50487	100	Min: 80 Max: 200	%	100	100
UAFE Max Torque Limit Adjust factor	50488	100	Min: 80 Max: 200	%	100	100
Accelerator Control						
Variable Rate Vehicle Acceleration Management	50040	N		Y = Enable N = Disable	Y	Y
Variable Acceleration 1	50041	0.6	Min: 0.0 Max: 8.1	mph/sec	1.5 kph/sec (0.93 mph/s)	1.5 kph/sec (0.93 mph/s)
Variable Acceleration 2	50042	0.6	Min: 0.0 Max: 8.1	mph/sec	1.5 kph/sec (0.93 mph/s)	1.5 kph/sec (0.93 mph/s)
Variable Acceleration Speed 1	50043	6	Min: 0 Max: 124	mph	20 kmph (12.43 mph)	20 kmph (12.43 mph)
Variable Acceleration Speed 2	50044	37	Min: 0 Max: 124	mph	60 kmph (37.28 mph)	60 kmph (37.28 mph)
Engine Brakes						
Engine Brake Control Enable	23021	N		Y = Enable N = Disable	Y	Y
Engine Brake Type	12790	0		0 = Exhaust Brake 2 = Engine Brake	2	2
Engine Brake Minimum Vehicle Speed	11540	0	Min: 0 Max: 35	mph	0	0
Engine Brake Delay Time	11542	0.0	Min: 0.0 Max: 4095.9	Seconds	0	0
Engine Brake Service Brake Activation Enable	21251	N		Y = Enable N = Disable	N	N
Engine Brake Clutch Pedal Activation Enable	21253	N		Y = Enable N = Disable	Y	Y
Cruise Control and Engine Brake Interaction Enable	20601	N		Y = Enable N = Disable	N	N
Start Maximum Engine Braking in Cruise Control	12410	3.1	Min: 0.0 Max: 6.2	mph	--	--
Start Minimum Engine Braking in Cruise Control	12420	2.0	Min: 0.0 Max: 6.2	mph	--	--
Accelerator Base						
Clutch Switch	10790	0		0 = Not Installed 1 = Installed	1	1
Starter Lockout Enable	24000	N		Y = Enable N = Disable	Y	Y
Starter Lockout Relay Type	22161	0		0 = Normally Closed 1 = Normally Open	1	1
Transmission Setup						
Transmission Top Gear Ratio	10300	1.000	Min: 0.000 Max: 64.000	Ratio	1	1
Gear Down Transmission Ratio	11340	1.330	Min: 0.000 Max: 64.000	Ratio	1.33	1.33
Transmission Type	11410	0		0 = Manual ; 1 = Automatic ; 2 = Fully Automated with Clutch ; 3 = Partially Automated 4 = Fully Automated without Clutch	0	0
Cruise Control Input						
Cruise Control	20021			Y = Enable N = Disable	Y	Y
Cruise control lower droop	10880			mph	0 kmph (0 mph)	0 kmph (0 mph)
Cruise control upper droop	10870			mph	0 kmph (0 mph)	0 kmph (0 mph)
Cruise control maximum vehicle speed	10090			mph	80 kmph (49.71 mph)	80 kmph (49.71 mph)
Cruise control auto resume	20611	N		Y = Enable N = Disable	N	N
Cruise control set speed save	21491	Y		Y = Enable N = Disable	Y	Y
Cruise Control and Engine Brake Interaction	20601	N		Y = Enable N = Disable	N	N
Switch Usage	10100	1		mph	1	1

Annexure 6

Vehicle Speed Input						
Rear Axle Ratio	10050	5.120	Min: 1.000 Max: 15.996		5.12	5.12
Tire Size	10060	500	Min: 301 Max: 1287	Rev/Mile	295 rev/km (475 rev/mile)	295 rev/km (475 rev/mile)
Number of Transmission Tailshaft Gear Teeth	10070	4	Min: 1 Max: 64		4	4
Vehicle Speed Sensor Type	10080	0		(0 = None) ; 1 = Magnetic; 3 = Data Link - Tailshaft	1	1
Maintenance Monitor						
Maintenance Monitor Enable	20131	N		Y = Enable N = Disable	Y	Y
Maintenance Monitor Alert Percentage	10860	90	Min: 0 Max: 100	%	90	90
Maintenance Monitor Operating Mode	10230	0	0=Automatic ; 1=Time 2=Fuel ; 3=Distance	0 = Automatic ; 1 = Time 2 = Fuel ; 3 = Distance	3	3
Maintenance Monitor Time	10290				1000 sec	1000 sec
Maintenance Monitor Distance	10280	25000	Min: 0 Max: 1000000	Miles	40000 km (24854.848 miles)	40000 km (24854.848 miles)
Maintenance Monitor Fuel	10250				2404	2404
Alternator Failure Warning	22571	N		Y = Enable N = Disable	Y	Y
Anti-Tampering Measures						
Maximum Engine Speed Without Vehicle Speed Source	10020	2850	Min: 900 Max: 3000	rpm	1850	1850
Vehicle Speed Sensor Anti-Tampering Enable	20761	N		Y = Enable N = Disable	Y	Y
Engine Protection						
Engine Protection Shutdown Enable	20091	Y		Y = Enable N = Disable	Y	Y
Service Brake						
Service Brake Switch Enable	50070	N		Y = Enable N = Disable	Y	Y
Fan Control						
Air Conditioning Pressure Switch Input	21712	Disable		Y = Enable N = Disable	Y	N
Min Fan ON Time for Air Conditioning Pressure Switch	11350			Seconds	5	...
Fan Type	12490				7	7
Fan drive ratio	50670			ratio	1.2	1.2
Air Condition Control (Applicable only for SIGNA cabin)						
Air Conditioner Control	51260				Y	N
Air Conditioner On Idle Speed Adjust	51262				N	...
Powertrain Protection (PTP)						
Powertrain Protection	21781	Disable		Y = Enable N = Disable	N	N